Aerospace Science and Technology 132 (2023) 107985

Contents lists available at ScienceDirect

Aerospace Science and Technology

journal homepage: www.elsevier.com/locate/aescte

Adaptive fault-tolerant controller for morphing aircraft based on the L2 gain and a neural network

LaoHu Yuan [a], LiDong Wang [a], JiangTao Xu [b,*]

[a] College of Aerospace Engineering, Shenyang Aerospace University, Shenyang, 110136, China
[b] College of Aerospace and Civil Engineering, Harbin Engineering University, Harbin, 150006, China

ARTICLE INFO

Article history:
Received 13 July 2022
Received in revised form 30 October 2022
Accepted 2 November 2022
Available online 16 November 2022
Communicated by Chaoyong Li

Keywords:
Morphing aircraft
Nonlinear aerodynamic modeling
Fault-tolerant controller
Attitude control
Recursive damped least squares

ABSTRACT

This paper presents an adaptive neural network fault-tolerant controller approach for the parameters of morphing aircraft that change violently during changing sweep angle processes and with morphing mechanisms that are prone to failure. We establish a nonlinear dynamic model of a morphing aircraft and analyze the variation laws of the aircraft parameters at different sweep angles. Considering the dynamic changes of the aircraft in morphing and morphing mechanism failure, by using recursive damped least squares (RDLS) to estimate the aircraft system model and offset the adverse effects of parameter mutation on the system, a radial basis function neural network (RBFNN) is used to further compensate for unmodeled parts such as actuator delay. The L2 gain is used to design a robust controller to constrain the errors generated by the RDLS and RBFNN in the learning process. The closed-loop stability is guaranteed by Lyapunov stability theory and the technique of compact set. Lastly, the comparison results of simulation indicate that this method satisfies the design criteria in the morphing process and fault situations, confirming the robustness of the controller.

© 2022 Published by Elsevier Masson SAS.

1. Introduction

Morphing aircrafts attract extensive attention from scholars worldwide because they have a flight shape that can change according to different mission environments [1] [2]. There are a variety of morphing modes, such as changing the aircraft flap [3], changing the wing span, and changing sweep angle [4] [5]. However, the morphing process causes sudden changes in aerodynamic parameters, the position of the center of gravity, and the moment of inertia. These changes make it difficult to design flight control systems [6] [7].

There are two main subfields of research in morphing aircraft attitude control. One uses the morphing signal as an external command, and the controller design is based only on the morphing aircraft dynamic model in following the command, which is independent of a feedback loop, the other uses the morphing mechanism to replace the control surface in the feedback loop. The method adopted depends on the structure and operating conditions of the aircraft, and most morphing aircraft adopt the first method. When morphing aircraft control systems are designed, the dynamic model is usually linearized [8] [9]. The switching linear variable parameter (LPV) method is a representative method. In [10], an LPV model was established using scheduled parameter interval division and convex decomposition. Considering the switching logic of the average dwell time (ADT) and mode-dependent average dwell time (MDADT), the gain-scheduling output feedback control problem of full-envelope morphing aircraft has been studied. Considering the uncertainty of parameters and asynchronous switching in LPV models, [11] proposed a nonfragile switched H_∞ controller, which overcomes the influence of dynamic model uncertainty. In [12], the linear fractional form (ILR) was utilized to develop a sliding mode controller for an LPV aircraft model with variable wingspan and variable sweep angle, and fuzzy gain adjustment was employed to compensate for the uncertainty. However, the above methods often ignore the nonlinear characteristics in the dynamic modeling process of deformable aircraft when constructing LPV models and cannot completely express the whole dynamic morphing aircraft process.

A nonlinear model can be used to create a controller, ensuring model accuracy. For the morphing aircraft to be able to adapt to a wider flying envelope, a switching controller was developed based on a nonlinear model in [13], taking into account changes in aerodynamic

* Corresponding author.
E-mail address: xujiangtao@hrbeu.edu.cn (J. Xu).

https://doi.org/10.1016/j.ast.2022.107985
1270-9638/© 2022 Published by Elsevier Masson SAS.

properties at various altitudes and Mach numbers. Nevertheless, the target altitude and the derivative of the target altitude need to be in a conservative compact set and cannot track a rapidly changing altitude command. Additionally, a radial basis function neural network (RBFNN) was used to approximate the dynamic model [14], and then a dynamic surface method was used to design an adaptive controller. In [15], an integrated guidance, control, and morphing (IGCM) method for a hypersonic variable vehicle (HMV) was designed using an adaptive block dynamic surface backstepping method. Unfortunately, the dynamic models of the vehicle need to be known completely during controller design. For a model-free control problem, [16] used deep reinforcement learning to control the autonomous morphing of an aircraft wing, taking the span length as the control input to control the aircraft attitude, but there was a large difference in angle tracking.

The complexity of the damage and ease of damaging the morphing mechanism, in addition to the challenge of controller design, is a significant barrier to its use for engineering. When designing morphing aircraft fault-tolerant controllers, some traditional aircraft fault-tolerant control methods can be used for reference [17] [18] [19]. Generally, faults can be divided into sensor faults [20], actuator faults [21] [22] [23], and system faults. System faults include loss of thrust (LOT) [24], wing damage, and the stability surface damage[25] [26]. The structure of a morphing aircraft mechanism is complex and is more prone to actuator failure of the mechanism in the flight process, resulting in system failure. At present, there are many methods of aircraft fault-tolerant control, which can be divided into two categories. One is fault-tolerant control based on hardware redundancy. This kind of control usually connects several identical components in parallel and switches to standby components when faults are found in the current component detection. The other is analytical redundant fault tolerance control, which is easier to implement in engineering. When an aircraft fails, it can perform a task or maintain performance only by allocating the remaining modules. This type of method can also be divided into two categories: active control and passive control.

Passive control takes the information generated during the fault into account when proposing a control law. Methods in this category are robust. An adaptive fault-tolerant controller based on a fuzzy neural network and a nonsingular fast terminal sliding-mode control scheme for a flexible flapping aircraft were presented to solve the problem of system uncertainty and actuator failure [27]. In [28], a fault-tolerant controller was designed with a critic-actor structure based on a neural network backstepping method. Based on integral reinforcement learning, an adaptive model-free fault-tolerant control scheme for a highly flexible aircraft with actuator faults [29]. In [30], incremental adaptive sliding mode control was used for vertical take-off and landing (VTOL), and it could provide effective control after single rotor failure. Moreover, an adaptive controller using engine failure state has been employed in a tailstock aircraft for attitude and position control [31]. In order to effectively design a passive fault-tolerant controller, it is usually necessary to know the approximate range of aircraft faults in advance. Consequently, achieving a good control effect for an unknown fault is difficult.

Active control usually diagnoses and isolates or estimates a fault module, feeds back the fault information to the controller for adjustment, or changes the reference input according to the fault. As shown in [32], this method can optimize the performance of fault detection, and a recovery method based on recoverable sets was proposed. For fault-tolerant tracking control of hybrid wing aircraft with unknown disturbances and actuator failures, an adaptive incremental nonlinear backstepping sliding mode controller was presented in [33], which was combined with an RBFNN observer to enhance the ability to suppress disturbances. Based on an accurate damage flight dynamics model, [34] presented a composite method of a linear quadratic regulator (LQR) and model reference adaptive control. [35] proposed a sliding mode control algorithm for the redistribution of the control law for aircraft with redundant actuators, and actuator faults in aircraft were identified by recursive least squares (RLS). When an actuator of a flexible aircraft fails completely, as in [36], the estimation method of an adaptive observer is used to estimate the uncertainty caused by flexibility, and a fault-tolerant controller was designed on this basis. An active fault-tolerant controller can deal with a variety of aircraft faults, but it is more sensitive to the aircraft model. Unfortunately, a mathematical model of an aircraft is often difficult to obtain accurately.

This study uses a method of backstepping to design an adaptive passive fault-tolerant controller based on the L_2 gain. Specifically, we establish a nonlinear dynamic model of morphing aircraft to analyze the aerodynamic characteristics. This model can be described accurately by comparing it with LPVs. An attitude adaptive robust attitude controller is designed using the L_2 gain. The contribution of this paper is that we combine a neural network and recursive damped least squares (RDLS) to respectively compensate for sudden changes in aerodynamic parameters and inertia moments caused by morphing and the unmodeled parts (such as rudder delay), which can greatly improve the robustness of this controller. On this basis, the controller can still track the rapidly changing attitude angle command when the fuselage is deformed and when the deformation mechanism fails.

The article is organized as follows: Section 2 analyzes the dynamic model and aerodynamic parameters of a morphing aircraft. Section 3 gives the designs of the fault-tolerant controller and a proof of its stability. Sections 4 and 5 respectively provide a contrast simulation analysis and the conclusions of this paper.

2. Morphing aircraft mathematical model

2.1. Morphing aircraft description

In this study, the morphing aircraft adopts the shape shown in Fig. 1, and the sweep angle can change between 0° and 30°. Because the aircraft is small, the wing mass and moment of inertia cannot be ignored. To facilitate modeling, in this study, we divide the aircraft into three parts, an airframe and two wings, and select the fuselage centroid as the origin of the body coordinate system. The elastic deformation effects of the morphing aircraft and its moving parts are ignored during the entire flight, and the aircraft wing and fuselage are simplified as a hinge connection.

Step 1: Translational equations of motion

The relationship between the external forces and the ground speed vector of the aircraft, that is, the centroid speed of the airframe relative to the speed $\boldsymbol{V}_g$ of the inertial frame, is as follows:

$$\boldsymbol{F} = m\frac{d\boldsymbol{V}_g}{dt} \tag{1}$$

where m is the total mass of the aircraft, and $\boldsymbol{F}$ is the sum of all external forces on the aircraft.

The center of gravity of the total aircraft wing position $\boldsymbol{r}_w$ in the body frame is:

$$\boldsymbol{r}_w = \frac{\sum_{i=1}^{n} m_i \boldsymbol{r}_i}{\sum_{i=1}^{n} m_i} \tag{2}$$

The velocity $\boldsymbol{V}_w$ of the C.G. of the total wing relative to the inertial coordinate systems is formulated as:

$$\boldsymbol{V}_w = \boldsymbol{V}_0 + \frac{d\boldsymbol{r}_w}{dt} = \boldsymbol{V}_0 + \dot{\boldsymbol{r}}_w + \boldsymbol{w} \times \boldsymbol{r}_w \tag{3}$$

where m_i is the mass of each aircraft wing, r_i is the distance vector from the C.G. of each wing to the C.G. of the airframe, $\boldsymbol{V}_0 = \begin{bmatrix} u & v & w \end{bmatrix}^T$ is the inertial velocity of the airframe C.G. in the body frame, and $\boldsymbol{w} = \begin{bmatrix} \mathrm{p} & q & r \end{bmatrix}^T$ is the angular rate of the fuselage in the body frame.

The defined ground speed vector $\boldsymbol{V}_g$ can be expressed as:

$$\boldsymbol{V}_g = \frac{1}{m}\left(m - \sum_{i=1}^{n} m_i\right)\boldsymbol{V}_0 + \frac{1}{m}\sum_{i=1}^{n} m_i \boldsymbol{V}_w \tag{4}$$

Eq. (4) can be expressed as:

$$\frac{1}{m}\boldsymbol{F} = \frac{d\boldsymbol{V}_g}{dt} = \dot{\boldsymbol{V}}_0 + \boldsymbol{w} \times \boldsymbol{V}_0 + \sum_{i=1}^{n} \frac{m_i}{m}\left(\dot{\boldsymbol{w}} \times \boldsymbol{r}_i + 2\boldsymbol{w} \times \dot{\boldsymbol{r}}_i + \boldsymbol{w} \times \boldsymbol{w} \times \boldsymbol{r}_i\right) \tag{5}$$

Most importantly, we obtain the translational equation of the aircraft during variant motion in the body coordinate system.

Step 2: Rotational equations of motion

Similarly, in the body frame, the moment of momentum is as follows:

$$\boldsymbol{M} = \frac{d\boldsymbol{H}}{dt} = \dot{\boldsymbol{H}} + \boldsymbol{w} \times \boldsymbol{H} \tag{6}$$

where $\boldsymbol{M}$ is the sum of all external moments on the aircraft, and $\boldsymbol{H}$ is the angular momentum of the aircraft, which is defined as the product of the total inertia matrix of the aircraft $\boldsymbol{I}$ and the angular rate vector:

$$\boldsymbol{H} = \boldsymbol{I}\boldsymbol{w} + \sum_{i=1}^{n}\left(m_i \boldsymbol{r}_i \times \boldsymbol{V}_g\right) \tag{7}$$

Eq. (7) can be expressed as:

$$\boldsymbol{M} = \dot{\boldsymbol{I}}\boldsymbol{w} + \boldsymbol{I}\dot{\boldsymbol{w}} + \boldsymbol{w} \times (\boldsymbol{I}\boldsymbol{w}) + \sum_{i=1}^{n}\left(m_i \boldsymbol{r}_i \times \mathrm{d}\boldsymbol{V}_g + m_i \boldsymbol{r}_i \times \left(\boldsymbol{w} \times \boldsymbol{V}_g\right) + m_i \dot{\boldsymbol{r}}_i \times \boldsymbol{V}_g\right) \tag{8}$$

where $\boldsymbol{I} = \boldsymbol{I}_b + \sum_{i=1}^{n}\left(\boldsymbol{A}_i \boldsymbol{I}_i \boldsymbol{A}_i^T + \boldsymbol{I}_{si}\right)$, $\boldsymbol{I}_b$ is the moment of inertia of the aircraft airframe, $\boldsymbol{I}_i$ is the moment of inertia of the wing relative to the connection point of the airframe and wing, $\boldsymbol{I}_{si}$ is the increment of the moment of inertia for the parallel axis theorem, and $\boldsymbol{A}_i$ is the transformation matrix from the coordinate system of the wing frame $o_i x_i y_i z_i$ to the coordinate system of the airframe $o_b x_b y_b z_b$. In Fig. 1 the wing frame z-axis is perpendicular to the surface of xoy and coincides with the body frame z-axis.

$A_i = \begin{bmatrix} \cos\delta_i & \sin\delta_i & 0 \\ -\sin\delta_i & \cos\delta_i & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and δ_i denote the rotation angle of the wing frame with respect to the body frame. In this study, $i = 1$ is defined as the right wing and $i = 2$ is defined the left wing.

Step 3: External forces $\boldsymbol{F}$ and external moments $\boldsymbol{M}$

$\boldsymbol{F}$ and $\boldsymbol{M}$ can be written as described in [33] [37]:

$$\boldsymbol{F} = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} \cos\beta\cos\alpha & -\sin\beta\cos\alpha & -\sin\alpha \\ \sin\beta & \cos\beta & 0 \\ \cos\beta\sin\alpha & -\sin\beta\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} -D \\ 0 \\ -L \end{bmatrix} + \begin{bmatrix} -\sin\theta \\ \cos\theta\sin\phi \\ \cos\theta\cos\phi \end{bmatrix} G + \begin{bmatrix} \cos\alpha_{motor} \\ 0 \\ \sin\alpha_{motor} \end{bmatrix} F_{motor} \tag{9}$$

$$\boldsymbol{M} = \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} l \\ m \\ n \end{bmatrix} \tag{10}$$

where α and β are the angle of attack and the side slip angle of the aircraft, respectively, α_{motor} is the engine installation angle, and F_{motor} is the engine thrust. The aircraft control surfaces including rudder, elevator, and aileron are limited to 0.52 [rad] (30°), and D, L, l, m, and n are the aerodynamic forces and moments of the aircraft.

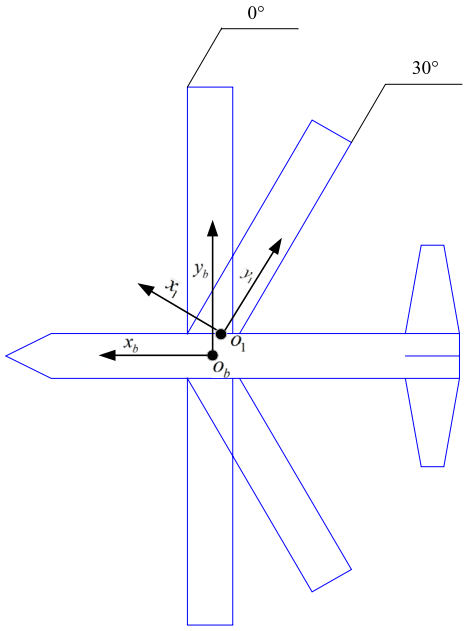


Fig. 1. Definition of axes and morphing.

2.2. Aircraft kinematic equation

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ \sin\theta & -\sin\phi\cos\theta & -\cos\phi\cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{11}$$

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \tag{12}$$

where ϕ, θ, and ψ are the roll angle, pitch angle, and yaw angle of the aircraft, respectively, and x, y, and h are the inertial positions of the aircraft.

2.3. Aerodynamic calculation and analysis

The morphing aircraft fly at low velocity, low altitude and low Reynolds number, we can assume that the aerodynamic parameters are independent of the flight altitude and velocity. The aerodynamic forces and moments can be described as follows:

$$\begin{aligned} L &= \tfrac{1}{2}\rho V_a^2 S\left(C_L(\alpha,\beta,\delta_i) + C_L^{\delta_e}\delta_e\right) \\ D &= \tfrac{1}{2}\rho V_a^2 S\left(C_D(\alpha,\beta,\delta_i) + C_D^{\delta_e}\delta_e\right) \\ m &= \tfrac{1}{2}\rho V_a^2 Sc\left(C_m(\alpha,\beta,\delta_i) + C_m^q\tfrac{c}{2V_a}q + C_m^{\delta_e}\delta_e\right) \\ l &= \tfrac{1}{2}\rho V_a^2 Sb\left(C_l(\alpha,\beta,\delta_i) + C_l^p\tfrac{b}{2V_a}p + C_l^r\tfrac{b}{2V_a}r + C_l^{\delta_a}\delta_a + C_l^{\delta_r}\delta_r\right) \\ n &= \tfrac{1}{2}\rho V_a^2 Sb\left(C_n(\alpha,\beta,\delta_i) + C_n^p\tfrac{b}{2V_a}p + C_n^r\tfrac{b}{2V_a}r + C_n^{\delta_a}\delta_a + C_n^{\delta_r}\delta_r\right) \end{aligned} \tag{13}$$

where ρ is the atmospheric density, V_a is the airspeed, S is the surface area of the aircraft wing, b is the wing span, c is the mean aerodynamic chord of the wing, δ_e, δ_a, and δ_r denote the control signals of elevator deflection, aileron deflection and rudder deflection, respectively, and C_i^j is the aerodynamic parameter of the aircraft, which in Eq. (13) is a dimensionless parameter.

There are many methods for calculating the aerodynamic parameters of an aircraft, such as wind tunnel experiments [38], CFD [39] [40], and datcom [41] [42]. We used XFLR5 software to calculate the aerodynamic data of the aircraft in this study, and in [43], identical wings were calculated using XFLR5 and a wind tunnel in a low-speed environment. The lift coefficient is in good agreement, and there is a small difference between the drag coefficient slope and pitching moment coefficient, which can meet the accuracy requirements for this study. We calculate the relationship between the aerodynamic force and aerodynamic moment coefficient with the angle of attack and sideslip angle when the sweep angle is 0 [rad] (0°), 0.1745 [rad] (10°), 0.3491 [rad] (20°), and 0.5236 [rad] (30°) and then use interpolation in dynamic modeling.

Fig. 2 shows the aerodynamic force and moment coefficients related to the angle of attack, sideslip angle, and sweep angle when the rudder control signal of the aircraft is 0 ($\delta_r = 0$, $\delta_a = 0$, $\delta_e = 0$). In XFLR5, the aerodynamic lateral force coefficient C_y is 0. For the longitudinal aerodynamic coefficient, the lift coefficient and drag coefficient of the aircraft decrease with increasing sweep angle. Generally, the lift-to-drag ratio of an aircraft decreases with increasing sweep angle. The change rate of the pitch moment coefficient with respect to the angle of attack decreases with increasing sweep angle because the effect of reducing the force arm is greater than that of reducing

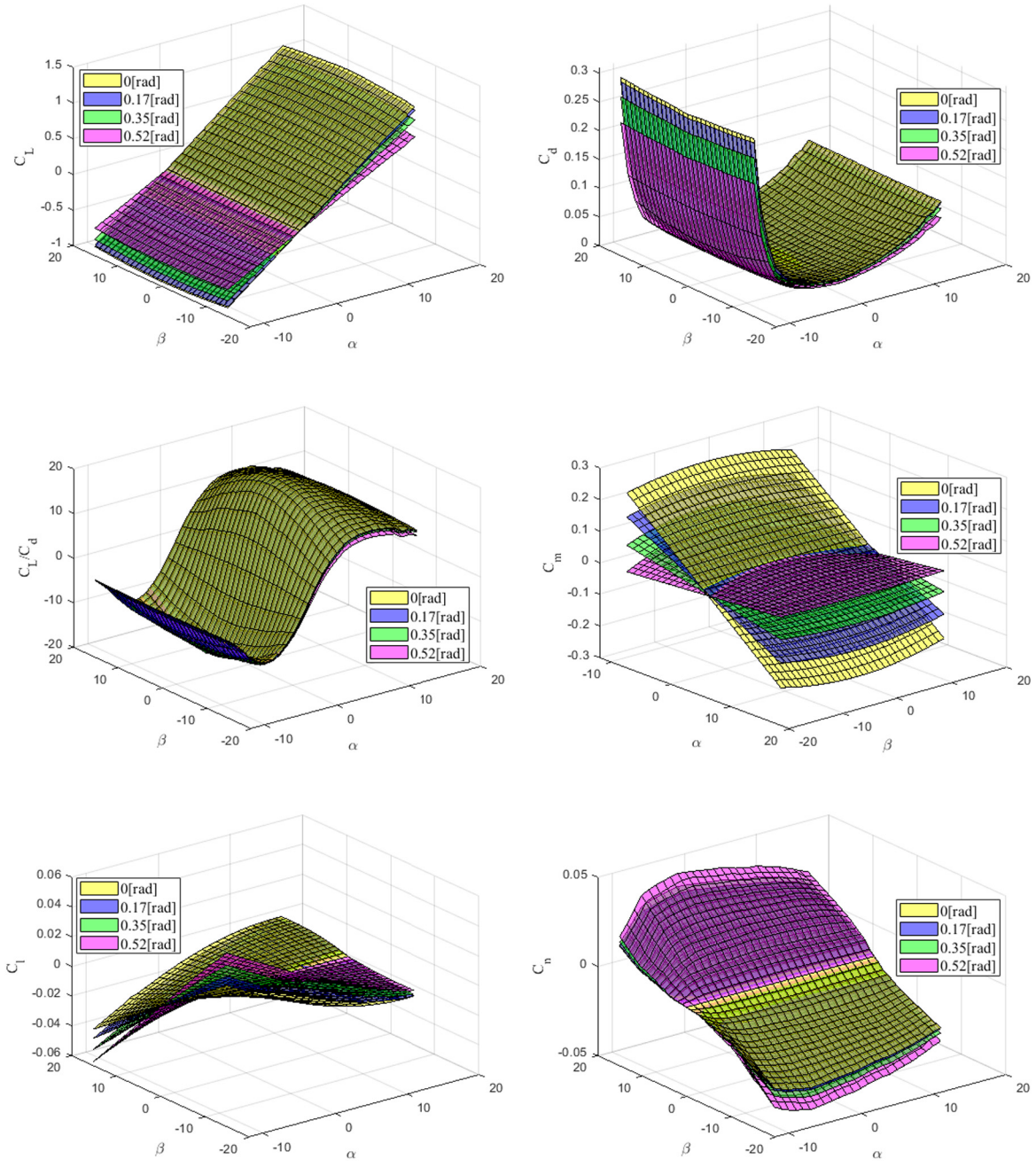


Fig. 2. Aerodynamic coefficient.

the lift. Regarding the transverse coefficient, the change rate of the yaw and roll moment coefficients with the side slip angle increases with increasing sweep angle, and the stability of the transverse heading is reduced by reducing the wingspan. Due to the small size of the aircraft compared with other aircraft, such as the model in [13], the change in the moment coefficient is more pronounced, which poses a great challenge for the design of the controller.

2.4. Actuator model

To properly simulate the aircraft flight state, mathematical models of the rudders and actuators with varying sweep angles are developed.

The actual model of the steering gear is a third-order nonlinear transfer function, which is equivalent to a first-order transfer function in this study [46]:

$$G_1 = \frac{37}{s + 37}$$

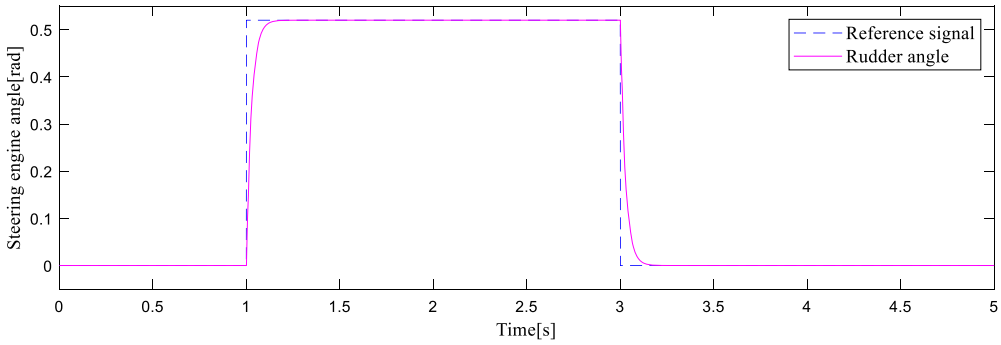


Fig. 3. Rudder response.

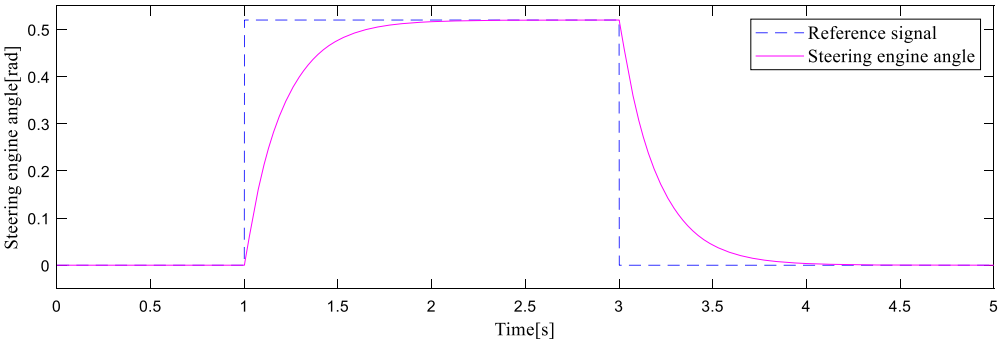


Fig. 4. Morphing mechanism response.

Table 1
UAV sweep angle and elevator deflection angle.

Sweep angle	δ_e
0 [rad]	−0.0543 [rad]
0.17 [rad]	−0.0455 [rad]
0.34 [rad]	−0.0335 [rad]
0.52 [rad]	−0.0126 [rad]

The transfer function of the actuator with varied sweep angles is defined as:

$$G_2 = \frac{5}{s+5}$$

The saturation and time-delay responses of the actuator will interfere with the control of the aircraft. The sweep angle of the actuator can be changed from 0 [rad] to 0.52 [rad] (30°) within 1 s, which prevents more drastic changes in aerodynamic parameters and external forces and moments caused by variable sweep.

The response curves of the rudder and the morphing mechanism of the variable sweep angle actuator are shown in Fig. 3 and Fig. 4, respectively.

2.5. Aircraft open-loop dynamic response

Step 1: Longitudinal open-loop response

Taking 80% of the maximum value of the throttle position as the throttle in straight-line horizontal flight, the sweep angle and elevator rudder angle in the trim are shown in Table 1.

Fig. 5 shows the longitudinal parameter changes of the aircraft. When the sweep angle increases and the angle of attack required to maintain horizontal flight also increases, the low head-pitching moment generated by the aerodynamic change is greater than that generated by the position offset of the wing center of gravity, which also makes the nonminimum phase characteristic of the pitch angle response increasingly obvious. In Fig. 2, the larger the sweep angle is, the smaller the drag coefficient, which leads to increased aircraft speed.

Step 2: Lateral open-loop response

After 5 s on the step 1 basis, a gust disturbance is added to the side of the airframe to verify the heading of the aircraft and the stability in the roll and yaw channels.

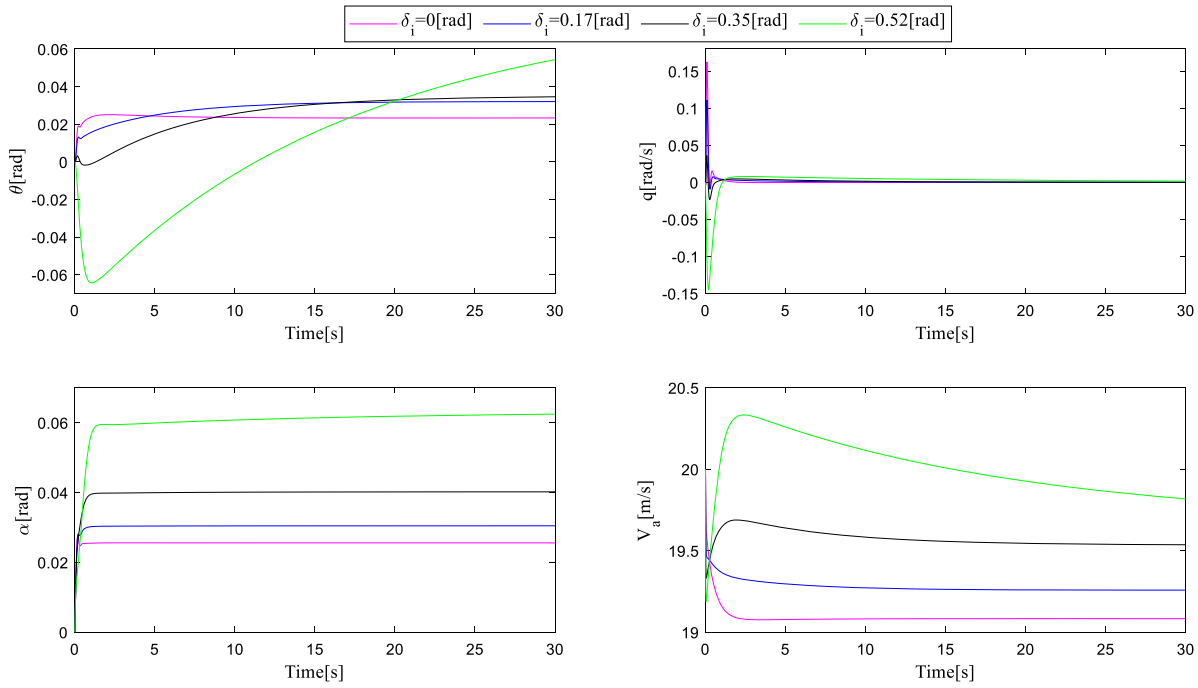


Fig. 5. Open loop dynamic responses in the lateral simulations.

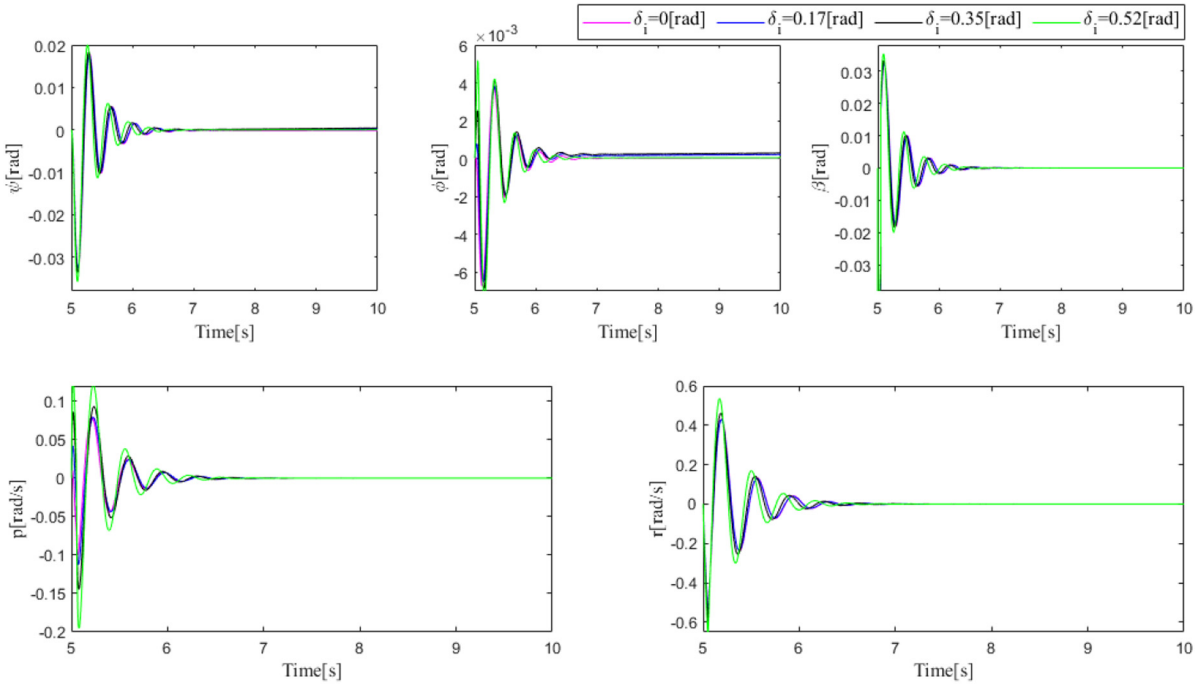


Fig. 6. Open loop dynamic responses in the longitudinal simulations.

In Fig. 6, with increasing sweep angle, the response amplitude of the aircraft also increases, while p and r in each state tend toward stability, and the short period parameter β also tends toward stability. However, there will be a certain static error, which will lead to a gradual increase in the long-period parameters ϕ and ψ.

3. Controller design

3.1. Assumptions and lemmas

Assumption 1. *When designing the controller, we can simplify Eq. (5) and Eq. (8) as the following equation:*

$$\begin{aligned} \dot{\boldsymbol{x}}_1 &= \boldsymbol{A}\left(\dot{\boldsymbol{x}}_1\right)\boldsymbol{x}_2 + \Delta_1 \\ \dot{\boldsymbol{x}}_2 &= \boldsymbol{f}\left(\boldsymbol{x}\right) + \boldsymbol{g}\boldsymbol{u} + \Delta_2 \end{aligned} \tag{14}$$

where $\boldsymbol{x}_2 = \begin{bmatrix} p & q & r \end{bmatrix}^T$, $\boldsymbol{x}_1 = \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^T$, Δ_2 *and* Δ_1 *are difficult to model dynamics such as the response delay and external disturbance of the pneumatic rudder.* $\boldsymbol{f}(\boldsymbol{x})$, $\boldsymbol{g}$ *and* $\boldsymbol{A}(\dot{\boldsymbol{x}}_1)$ *are provided in the Appendix.*

Lemma 1. *For the models:*

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}\boldsymbol{d}$$
$$\boldsymbol{z} = \boldsymbol{h}(\boldsymbol{x})$$

where $\boldsymbol{d} = \boldsymbol{d}(t)$*, which is an external interference, and the* L_2 *norm is* $\|\boldsymbol{d}(t)\|_2 = \left\{\int_0^\infty \boldsymbol{d}^T(t)\,\boldsymbol{d}(t)\,dt\right\}^{\frac{1}{2}}$*, which can measure the energy of external interference* $\boldsymbol{d}$*.* $\boldsymbol{z} = \boldsymbol{h}(\boldsymbol{x})$ *is the performance evaluation signal of the system., such as error, which are known continuous vector or matrix function of appropriate dimensions.*

We define the gain J that can be used to evaluate the ability of the model system to suppress external interference:

$$J = \sup_{\|\boldsymbol{d}\|_2 \neq 0} \frac{\|\boldsymbol{z}\|_2}{\|\boldsymbol{d}\|_2}$$

J can represent the robustness of the system. The greater J is, the worse the robustness of the system. In contrast, the robustness of the system improves with decreasing J.

Based on the above definition, Lemma 1 is given according to [44]:

Given a constant $\gamma > 0$, there is a differentiable function $V \geq 0$, that satisfies the following Hamilton Jacobi inequality:

$$\dot{V} = \frac{\partial V}{\partial \boldsymbol{x}}\dot{\boldsymbol{x}} = \frac{\partial V}{\partial \boldsymbol{x}}\boldsymbol{f}(\boldsymbol{x}) + \frac{\partial V}{\partial \boldsymbol{x}}\boldsymbol{g}(\boldsymbol{x})\,\boldsymbol{d} \leq \frac{1}{2}\left\{\gamma^2\|\boldsymbol{d}\|^2 - \|\boldsymbol{z}\|^2\right\}, \forall \boldsymbol{d} \tag{15}$$

Then, the performance index of the system $J \leq \gamma$.

Lemma 2. *A neural network is used to address the difficulty in modeling and compensating for the terms in Eq. (14) In this study, a radial basis function neural network (RBFNN) is used. According to [45]:*

For any function $F(\boldsymbol{x}_{in})$*, use RBFNN* $F_{NN}(\boldsymbol{x}_{in}) = \boldsymbol{w}\boldsymbol{\Pi}(\boldsymbol{x}_{in})$ *to approximate* $F(\boldsymbol{x}_{in})$*:*

$$F(\boldsymbol{x}_{in}) = \bar{\boldsymbol{w}}\boldsymbol{\Pi}(\boldsymbol{x}_{in}) = \boldsymbol{w}\boldsymbol{\Pi}(\boldsymbol{x}_{in}) + \boldsymbol{\varepsilon} \tag{16}$$

where $\bar{\boldsymbol{w}} = [\bar{w}_1 \ldots \bar{w}_n]$ *is the ideal weight value of the RBFNN, and* $\boldsymbol{w} = [w_1 \ldots w_n]$ *is the real weight value of the RBFNN.* $\boldsymbol{\Pi}(\boldsymbol{x}_{in}) = [\Pi_1(\boldsymbol{x}_{in}) \ldots \Pi_n(\boldsymbol{x}_{in})]^T$ $\boldsymbol{\Pi}(\boldsymbol{x}_{in}) : \mathbb{R}^n \rightarrow \mathbb{R}$ *is defined as the basis function of the neural network, which is a series of bound and integrable continuous functions.* $\Pi_i(\boldsymbol{x}_{in}) = \frac{\exp(-\|\boldsymbol{x}_{in} - c_i\|_2^2)}{b^2}$*,* $\boldsymbol{x}_{in}$ *is the RBFNN input.* c_i *is the RBFNN center, and b is the RBFNN width. The actual weight and error are bounded:* $\|\boldsymbol{w}\| \leq w_M$*,* $\|\boldsymbol{\varepsilon}\| < \varepsilon_M$*.*

3.2. Controller design

We imitate backstepping control to design a progressive controller.

Step 1: The error $\boldsymbol{e}_1 = \boldsymbol{x}_{1d} - \boldsymbol{x}_1$ is defined, where $\boldsymbol{x}_{1d} = \begin{bmatrix} \phi_d & \theta_d & \psi_d \end{bmatrix}^T$, ϕ_d, θ_d, and ψ_d are the target roll angle, target pitch angle and target yaw angle, respectively.

Taking the derivative of $\boldsymbol{e}_1$, we have:

$$\dot{\boldsymbol{e}}_1 = \dot{\boldsymbol{x}}_{1d} - \dot{\boldsymbol{x}}_1 = \dot{\boldsymbol{x}}_{1d} - \boldsymbol{A}(\dot{\boldsymbol{x}}_1)\,\boldsymbol{x}_2 - \Delta_1 \tag{17}$$

The virtual control law $\boldsymbol{x}_{2d}$ is defined as:

$$\boldsymbol{x}_{2d} = \boldsymbol{A}(\dot{\boldsymbol{x}}_1)^{-1}\left(k_1\boldsymbol{E}_1 - \Delta_1 + \frac{\boldsymbol{E}_1}{\gamma_1} + \dot{\boldsymbol{x}}_{1d}\right) \tag{18}$$

where $\boldsymbol{E}_1 = \boldsymbol{e}_1 + c_1\dot{\boldsymbol{e}}_1$, k_1 and γ_1 are robust controller parameters. The RBFNN is used to approximate Δ_1:

$$\Delta_1 = \bar{\boldsymbol{w}}_1\boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) = \boldsymbol{w}_1\boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) + \boldsymbol{\varepsilon}_1 \tag{19}$$

The weight value update rate of the RBFNN is $\dot{\bar{\boldsymbol{w}}}_1$:

$$\dot{\bar{\boldsymbol{w}}}_1 = \eta_1\boldsymbol{x}_{in}\boldsymbol{\Pi}_1 + \eta_1\lambda_1\|\boldsymbol{x}_{in}\|_2\,\boldsymbol{w}_1 \tag{20}$$

$\boldsymbol{x}$ is the input to the neural network and is defined as $\boldsymbol{x}_{in} = k_1\boldsymbol{E}_1$. η_1 and λ_1 are the learning rate of RBFNN.

Step 2: The error $\boldsymbol{e}_2 = \boldsymbol{x}_{2d} - \boldsymbol{x}_2$ is defined, where $\boldsymbol{x}_{2d} = \begin{bmatrix} p_d & q_d & r_d \end{bmatrix}^T$, p_d, q_d, and r_d are the target roll angular rate, target pitch angular rate, and target yaw angular rate.

Taking the derivative of $\boldsymbol{e}_2$ yields:

$$\dot{\boldsymbol{e}}_2 = \dot{\boldsymbol{x}}_{2d} - \dot{\boldsymbol{x}}_2 = \dot{\boldsymbol{x}}_{2d} - (f(\boldsymbol{x}) + \boldsymbol{g}\boldsymbol{u}) - \Delta_2 \tag{21}$$

The control law $\boldsymbol{u}$ is defined as:

$$\boldsymbol{u} = \boldsymbol{g}^{-1}\left(\mathrm{k}_2\boldsymbol{E}_2 - \Delta_2 + \frac{\boldsymbol{E}_2}{\gamma_2} + f(\boldsymbol{x}) + \dot{\boldsymbol{x}}_{2d}\right) \tag{22}$$

where $\boldsymbol{E}_2 = \boldsymbol{e}_2 + c_2\dot{\boldsymbol{e}}_2$, k_2 and γ_2 are robust controller parameters. The RBFNN is used to approximate Δ_2:

$$\Delta_2 = \bar{\boldsymbol{w}}_2\boldsymbol{\Pi}_2(\boldsymbol{x}_{in}) = \boldsymbol{w}_2\boldsymbol{\Pi}_2(\boldsymbol{x}_{in}) + \boldsymbol{\varepsilon}_2 \tag{23}$$

The weight value update rate of the RBFNN is $\dot{\bar{\boldsymbol{w}}}_2$:

$$\dot{\bar{\boldsymbol{w}}}_2 = \eta_2\boldsymbol{x}_{in}\boldsymbol{\Pi}_2 + \eta_2\lambda_2\|\boldsymbol{x}_{in}\|_2\boldsymbol{w}_2 \tag{24}$$

$\boldsymbol{x}$ is the input to the neural network, which define as $\boldsymbol{x}_{in} = \mathrm{k}_2\boldsymbol{E}_2$, η_2 and λ_2 are the learning rates of the RBFNN.

Step 3: In Eq. (15), the nominal model of $\boldsymbol{f}(\boldsymbol{x})$ is known, and $\boldsymbol{f}(\boldsymbol{x})$ can be fitted by the recursive damped least square (RDLS) algorithm.

Fig. 4 shows that the aircraft sweep angle changes rapidly from 0 to 0.52 [rad] (30°) within 1 s, and we can ignore the influence of the aircraft variation. According to Fig. 2, in the range of a small angle of attack and small sideslip angle, the moment coefficient can be regarded as a linear superposition of the angle of attack and sideslip angle as follows:

$$\begin{aligned} C_l(\alpha,\beta,\delta_1) &= C_l^0 + C_l^\alpha\alpha + C_l^\beta\beta \\ C_m(\alpha,\beta,\delta_1) &= C_m^0 + C_m^\alpha\alpha + C_m^\beta\beta \\ C_n(\alpha,\beta,\delta_1) &= C_n^0 + C_n^\alpha\alpha + C_n^\beta\beta \end{aligned} \tag{25}$$

The lateral aerodynamic coefficient is coupled, and we can define C_p and C_r for convenient expression:

$$\begin{aligned} C_p &= \mathcal{T}_3 C_l + \mathcal{T}_4 C_n \\ C_r &= \mathcal{T}_4 C_l + \mathcal{T}_8 C_n \end{aligned} \tag{26}$$

where $\mathcal{T}_3 = I_{zz}/\mathcal{T}$, $\mathcal{T}_4 = I_{xz}/\mathcal{T}$, $\mathcal{T}_8 = I_{xx}/\mathcal{T}$, and $\mathcal{T} = I_{xx}I_{zz} - I_{xz}^2$.

According to Eq. (13), we can obtain:

$$C_p = C_p(\alpha,\beta,\delta_1) + C_p^p\frac{b}{2V_a}p + C_p^r\frac{b}{2V_a}r + C_p^{\delta_a}\delta_a + C_p^{\delta_r}\delta_r$$

$$C_r = C_r(\alpha,\beta,\delta_1) + C_r^p\frac{b}{2V_a}p + C_r^r\frac{b}{2V_a}r + C_r^{\delta_a}\delta_a + C_r^{\delta_r}\delta_r$$

In Eq. (14), $\dot{\boldsymbol{x}}_2$ can be expressed as follows:

$$\dot{\boldsymbol{x}}_2 = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}\boldsymbol{u} = \begin{bmatrix} \mathbf{H_1}\bar{\boldsymbol{\Xi}}_\mathbf{1} \\ \boldsymbol{H_2}\bar{\boldsymbol{\Xi}}_\mathbf{2} \\ \boldsymbol{H_3}\bar{\boldsymbol{\Xi}}_\mathbf{3} \end{bmatrix} \tag{27}$$

where:

$$\begin{aligned} \boldsymbol{H}_1 &= \begin{bmatrix} pq & qr & 1 & \alpha & \beta & p & r & \delta_a & \delta_r \end{bmatrix} \\ \boldsymbol{H}_2 &= \begin{bmatrix} (r^2 - p^2) & pr & 1 & \alpha & \beta & q & \delta_e \end{bmatrix} \\ \boldsymbol{H}_3 &= \begin{bmatrix} pq & qr & 1 & \alpha & \beta & p & r & \delta_a & \delta_r \end{bmatrix} \\ \bar{\boldsymbol{\Xi}}_\mathbf{1} &= \tfrac{1}{2}\rho V_a^2\begin{bmatrix} \frac{2\mathcal{T}_1}{\rho V_a^2} & -\frac{2\mathcal{T}_2}{\rho V_a^2}{}_2 & C_{p0} & C_p^\alpha & C_p^\beta & \frac{b}{2V_a}C_p^p & \frac{b}{2V_a}C_p^r & C_p^{\delta_a} & C_p^{\delta_r} \end{bmatrix}^T \\ \bar{\boldsymbol{\Xi}}_2 &= \tfrac{1}{2}\rho V_a^2\begin{bmatrix} \frac{2\mathcal{T}_5}{\rho V_a^2} & -\frac{2\mathcal{T}_6}{\rho V_a^2} & C_{m0} & C_m^\alpha & C_m^\beta & \frac{c}{2V_a}C_m^q & C_p^{\delta_e} \end{bmatrix}^T \\ \bar{\boldsymbol{\Xi}}_3 &= \tfrac{1}{2}\rho V_a^2\begin{bmatrix} \frac{2\mathcal{T}_7}{\rho V_a^2} & -\frac{2\mathcal{T}_1}{\rho V_a^2}{}_2 & C_{r0} & C_r^\alpha & C_r^\beta & \frac{b}{2V_a}C_r^p & \frac{b}{2V_a}C_r^r & C_r^{\delta_a} & C_r^{\delta_r} \end{bmatrix}^T \end{aligned}$$

$$\mathcal{T}_1 = I_{xz}\left(I_{xx} - I_{yy} + I_{zz}\right)/\mathcal{T}, \quad \mathcal{T}_2 = \left(I_{zz}\left(I_{zz} - I_{yy}\right) + I_{xz}^2\right)/\mathcal{T}, \quad \mathcal{T}_5 = \left(I_{zz} - I_{xx}\right)/I_{yy}, \quad \mathcal{T}_6 = I_{xz}/I_{yy},$$

$$\mathcal{T}_7 = \left(\left(I_{xx} - I_{yy}\right)I_{xx} + I_{xz}^2\right)/\mathcal{T}$$

The system can be identified as follows:

$$\boldsymbol{z}(k) = \boldsymbol{H}(k)\boldsymbol{\Xi} = \boldsymbol{H}(k)\bar{\boldsymbol{\Xi}} + \boldsymbol{\varepsilon}_3 \tag{28}$$

where $\boldsymbol{H}(k)$ is the input to the system, $\boldsymbol{z}(k)$ is the system output, $\boldsymbol{\Xi}$ is the real value of a parameter to be identified, $\bar{\boldsymbol{\Xi}}$ is its calculated value, and $\boldsymbol{\varepsilon}_3$ is the system noise. It is assumed that the noise is white noise with zero mean value. The RDLS formula is as follows:

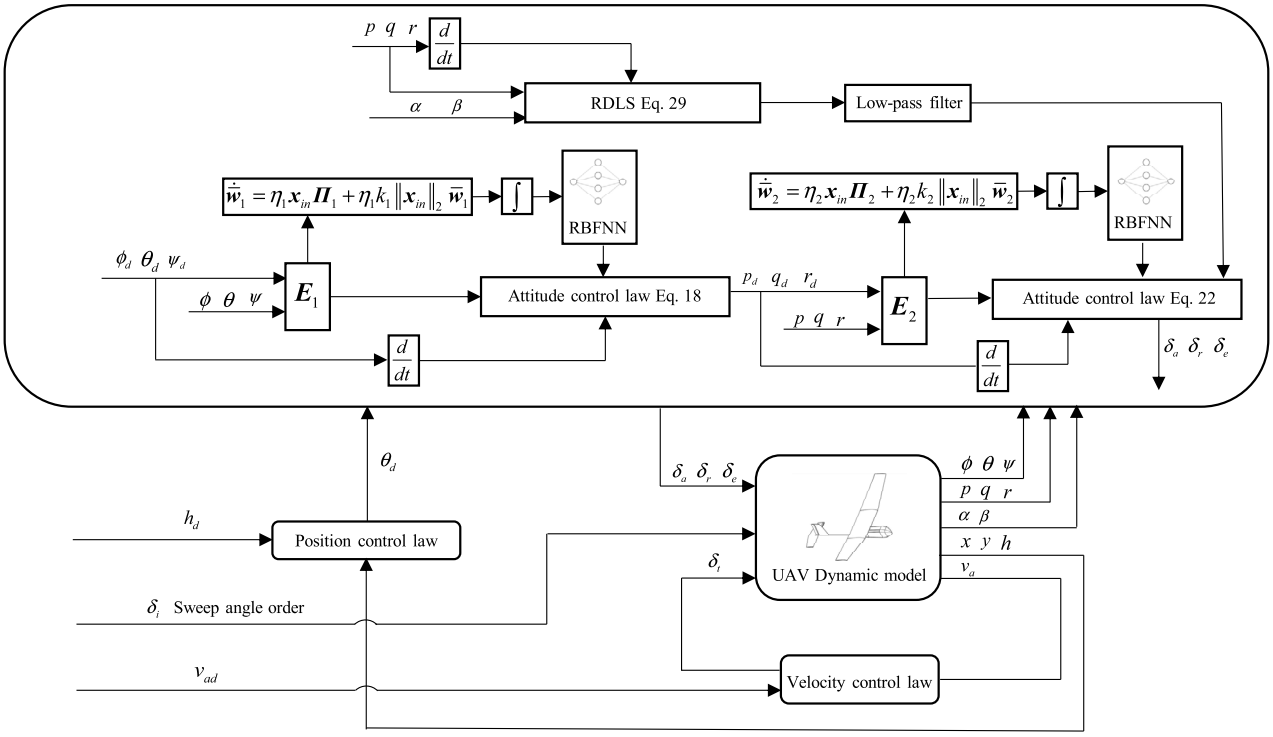


Fig. 7. Control scheme.

$$\begin{aligned}
&\boldsymbol{K}(k) = \boldsymbol{P}(k-1)\boldsymbol{H}(k)\left[\boldsymbol{H}^{\mathrm{T}}(k)\boldsymbol{P}(k-1)\boldsymbol{H}(k) + \tfrac{1}{\boldsymbol{\Lambda}(k)}\right]^{-1} \\
&\bar{\boldsymbol{\Xi}}(k) = \bar{\boldsymbol{\Xi}}(k-1) + \boldsymbol{K}(k)\left[\boldsymbol{z}(k) - \boldsymbol{H}^{\mathrm{T}}(k)\bar{\boldsymbol{\Xi}}(k-1)\right] \\
&\boldsymbol{P}(k) = \left[\boldsymbol{I} - \boldsymbol{K}(k)\boldsymbol{H}^{\mathrm{T}}(k)\right]\boldsymbol{P}(k-1)
\end{aligned} \tag{29}$$

The damping factor acts as an exponential window and weighted average for all data points so that the current data have a larger impact on the identified parameters, while the past data have a smaller impact on the identified parameters. When calculating the aircraft parameters online through the RDLS, it is necessary to ensure that the data in $\bar{\boldsymbol{\Xi}}$ are linearly independent. Taking $\bar{\boldsymbol{\Xi}}_{\mathbf{1}}$ as an example, the calculated moments of inertia $\mathcal{T}_1$, $\mathcal{T}_2$ and dynamic derivatives C_p^p, C_p^r will deviate, but the value of $\mathcal{T}_1 pq + \mathcal{T}_2 qr + C_p^p p + C_p^r r$ can be calculated.

The control scheme is shown in Fig. 7.

3.3. Proof of stability

According to Eqs. (17)–(19) and Eqs. (21)–(23), we obtain the following equations:

$$\begin{aligned}
\dot{\boldsymbol{e}}_1 &= \dot{\boldsymbol{x}}_{1d} - \boldsymbol{A}(\dot{\boldsymbol{x}}_1)\boldsymbol{x}_2 - \Delta_1 \\
&= \dot{\boldsymbol{x}}_{1d} - \boldsymbol{A}(\dot{\boldsymbol{x}}_1)(\boldsymbol{e}_2 + \boldsymbol{x}_{2d}) - \bar{\boldsymbol{w}}_1\boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) \\
&= \dot{\boldsymbol{x}}_{1d} - \boldsymbol{A}(\dot{\boldsymbol{x}}_1)\boldsymbol{e}_2 - k_1\boldsymbol{E}_1 + \boldsymbol{w}_1\boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) + \boldsymbol{\varepsilon}_1 - \tfrac{\boldsymbol{E}_1}{\gamma_1} - \dot{\boldsymbol{x}}_{1d} - \bar{\boldsymbol{w}}_1\boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) \\
&= -\boldsymbol{A}(\dot{\boldsymbol{x}}_1)\boldsymbol{e}_2 - k_1\boldsymbol{E}_1 - \tfrac{\boldsymbol{E}_1}{\gamma_1} - (\bar{\boldsymbol{w}}_1 - \boldsymbol{w}_1)\boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) + \boldsymbol{\varepsilon}_1
\end{aligned} \tag{30}$$

$$\begin{aligned}
\dot{\boldsymbol{e}}_2 &= \dot{\boldsymbol{x}}_{2d} - (\boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}\boldsymbol{u}) - \Delta_2 \\
&= \dot{\boldsymbol{x}}_{2d} - \boldsymbol{H}(k)\,\bar{\boldsymbol{\Xi}} - k_2\boldsymbol{E}_2 + \boldsymbol{w}_2\boldsymbol{\Pi}_2(\boldsymbol{x}_{in}) + \boldsymbol{\varepsilon}_2 - \tfrac{\boldsymbol{E}_2}{\gamma_2} + \boldsymbol{H}(k)\,\boldsymbol{\Xi} - \dot{\boldsymbol{x}}_{2d} - \bar{\boldsymbol{w}}_2\boldsymbol{\Pi}_2(\boldsymbol{x}_{in}) \\
&= -k_2\boldsymbol{E}_2 - \tfrac{\boldsymbol{E}_2}{\gamma_2} - (\bar{\boldsymbol{w}}_2 - \boldsymbol{w}_2)\boldsymbol{\Pi}_2(\boldsymbol{x}_{in}) + \boldsymbol{\varepsilon}_2 - \boldsymbol{\varepsilon}_3
\end{aligned} \tag{31}$$

Taking the derivatives of Eqs. (30) and (31) yields:

$$\ddot{\boldsymbol{e}}_1 = -\boldsymbol{A}(\dot{\boldsymbol{x}}_1)\dot{\boldsymbol{e}}_2 - k_1\dot{\boldsymbol{E}}_1 - \frac{\dot{\boldsymbol{E}}_1}{\gamma_1} \tag{32}$$

$$\ddot{\boldsymbol{e}}_2 = -k_2\dot{\boldsymbol{E}}_2 - \frac{\dot{\boldsymbol{E}}_2}{\gamma_2} \tag{33}$$

According to Eqs. (30) and (32), we can obtain $\dot{\boldsymbol{E}}_1$

$$\begin{aligned}
\dot{\boldsymbol{E}}_1 &= \dot{\boldsymbol{e}}_1 + c_1\ddot{\boldsymbol{e}}_1 \\
&= -\left(\boldsymbol{A}(\dot{\boldsymbol{x}}_1)\boldsymbol{e}_2 + k_1\boldsymbol{E}_1 + \tfrac{\boldsymbol{E}_1}{\gamma_1} + \boldsymbol{M}_1 + c_1 A\dot{\boldsymbol{e}}_2 + c_1 k_1\dot{\boldsymbol{E}}_1 + c_1\tfrac{\dot{\boldsymbol{E}}_1}{\gamma_1}\right) \\
&= -\left(\boldsymbol{A}(\dot{\boldsymbol{x}}_1)(\boldsymbol{e}_2 + c_1\dot{\boldsymbol{e}}_2) + \left(k_1 + \tfrac{1}{\gamma_1}\right)\boldsymbol{E}_1 + c_1\left(k_1 + \tfrac{1}{\gamma_1}\right)\dot{\boldsymbol{E}}_1 + \boldsymbol{M}_1\right) \\
&= m_1\boldsymbol{E}_1 + \boldsymbol{m}_2\boldsymbol{E}_2 + m_3\boldsymbol{M}_1
\end{aligned} \tag{34}$$

where $\boldsymbol{M}_1 = -(\bar{\boldsymbol{w}}_1 - \boldsymbol{w}_1)\boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) - \boldsymbol{\varepsilon}_1$, $m_1 = -(\gamma_1 k_1 + 1)/\left(\gamma_1^2 + \gamma_1 c_1 k_1 + 1\right)$, $\boldsymbol{m}_2 = -\gamma_1 \boldsymbol{A}/(\gamma_1 + \gamma_1 c_1 k_1 + 1)$, and $m_3 = -\gamma_1/(\gamma_1 + \gamma_1 c_1 k_1 + 1)$.

Similarly, we can obtain $\dot{\boldsymbol{E}}_2$

$$\dot{\boldsymbol{E}}_2 = m_4 \boldsymbol{E}_2 + m_5 \boldsymbol{M}_2 \tag{35}$$

where $\boldsymbol{M}_2 = -(\bar{\boldsymbol{w}}_2 - \boldsymbol{w}_2)\boldsymbol{\Pi}_2(x) - \boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3$, $m_4 = -(\gamma_2 k_2 + 1)/(\gamma_2 + \gamma_2 c_2 k_2 + 1)$, and $m_5 = -\gamma_2/(\gamma_2 + \gamma_2 c_2 k_2 + 1)$.

The Lyapunov function is selected:

$$V = V_1 + V_2 \tag{36}$$

where $V_1 = \frac{1}{2}\left(\boldsymbol{E}_1^T \boldsymbol{E}_1 + \boldsymbol{E}_2^T \boldsymbol{E}_2\right)$, $V_2 = \frac{1}{2} tr\left(\hat{\boldsymbol{w}}_1 \hat{\boldsymbol{w}}_1^T\right) + \frac{1}{2} tr\left(\hat{\boldsymbol{w}}_2 \hat{\boldsymbol{w}}_2^T\right)$, and V_1, V_2 are scalars.

Taking the derivative of V_1 yields:

$$\begin{aligned}
\dot{V}_1 &= \frac{1}{2}\left(\dot{\boldsymbol{E}}_1^T \boldsymbol{E}_1 + \boldsymbol{E}_1^T \dot{\boldsymbol{E}}_1 + \dot{\boldsymbol{E}}_2^T \boldsymbol{E}_2 + \boldsymbol{E}_2^T \dot{\boldsymbol{E}}_2\right) \\
&= \frac{1}{2}\left(\begin{array}{l} \left(m_1 \boldsymbol{E}_1 + \boldsymbol{m}_2(\boldsymbol{e}_2 + c_1 \dot{\boldsymbol{e}}_2) + m_3 \boldsymbol{M}_1\right)^T \boldsymbol{E}_1 + \boldsymbol{E}_1^T \left(m_1 \boldsymbol{E}_1 + \boldsymbol{m}_2(\boldsymbol{e}_2 + c_1 \dot{\boldsymbol{e}}_2) + m_3 \boldsymbol{M}_1\right) \\ + \left(m_4 \boldsymbol{E}_2 + m_5 \boldsymbol{M}_2\right)^T \boldsymbol{E}_2 + \boldsymbol{E}_2^T \left(m_4 \boldsymbol{E}_2 + m_5 \boldsymbol{M}_2\right) \end{array}\right) \\
&= m_1 \boldsymbol{E}_1^T \boldsymbol{E}_1 + m_4 \boldsymbol{E}_2^T \boldsymbol{E}_2 + m_2 \hat{A} \boldsymbol{E}_2^T \boldsymbol{E}_1 + m_3 \boldsymbol{M}_1^T \boldsymbol{E}_1 + m_5 \boldsymbol{M}_2^T \boldsymbol{E}_2
\end{aligned} \tag{37}$$

where $\hat{A} = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}$, a_{ij} is the element of matrix $\boldsymbol{A}(\dot{\boldsymbol{x}}_1)$.

Taking the derivative of V_2 yields:

$$\dot{V}_2 = \mathrm{tr}\left(\eta_1 \boldsymbol{E}_1 \boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) + \eta_1 \lambda_1 \|\boldsymbol{E}_1\|_2 \bar{\boldsymbol{w}}_1\right) \hat{\boldsymbol{w}}_1^T + tr\left(\eta_2 \boldsymbol{E}_2 \boldsymbol{\Pi}_2(\boldsymbol{x}_{in}) + \eta_2 \lambda_2 \|\boldsymbol{E}_2\|_2 \boldsymbol{w}_2\right) \hat{\boldsymbol{w}}_2^T \tag{38}$$

According to Eqs. (37) and (38), the derivative of V is:

$$\begin{aligned}
\dot{V} &= m_1 \boldsymbol{E}_1^T \boldsymbol{E}_1 + m_4 \boldsymbol{E}_2^T \boldsymbol{E}_2 + m_2 \hat{A} \boldsymbol{E}_2^T \boldsymbol{E}_1 + m_3 \left(\hat{\boldsymbol{w}}_1 \boldsymbol{\Pi}_1(\boldsymbol{x}_{in}) - \boldsymbol{\varepsilon}_1\right)^T \boldsymbol{E}_1 + m_5 \left(\hat{\boldsymbol{w}}_2 \boldsymbol{\Pi}_2(\boldsymbol{x}_{in}) - \boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\right)^T \boldsymbol{E}_2 \\
&\quad + tr\left(\eta_1 \boldsymbol{E}_1 \boldsymbol{\Pi}_1^T(\boldsymbol{x}_{in}) \hat{\boldsymbol{w}}_1^T + \eta_1 \lambda_1 \|\boldsymbol{E}_1\|_2 \bar{\boldsymbol{w}}_1 \hat{\boldsymbol{w}}_1^T\right) + tr\left(\eta_2 \boldsymbol{E}_2 \boldsymbol{\Pi}_2^T(\boldsymbol{x}_{in}) \hat{\boldsymbol{w}}_2^T + \eta_2 \lambda_2 \|\boldsymbol{E}_2\|_2 \bar{\boldsymbol{w}}_2 \hat{\boldsymbol{w}}_2^T\right)
\end{aligned} \tag{39}$$

We can obtain:

$$\boldsymbol{\Pi}_1^T(\boldsymbol{x}_{in}) \hat{\boldsymbol{w}}_1^T \boldsymbol{E}_1 = tr\left(\boldsymbol{E}_1 \boldsymbol{\Pi}_1^T(\boldsymbol{x}_{in}) \hat{\boldsymbol{w}}_1^T\right)$$

$$\boldsymbol{\Pi}_2^T(\boldsymbol{x}_{in}) \hat{\boldsymbol{w}}_2^T \boldsymbol{E}_2 = tr\left(\boldsymbol{E}_2 \boldsymbol{\Pi}_2^T(\boldsymbol{x}_{in}) \hat{\boldsymbol{w}}_2^T\right)$$

$$\boldsymbol{E}_2^T \boldsymbol{E}_1 \le \frac{1}{2}\left(\boldsymbol{E}_2^T \boldsymbol{E}_2 + \boldsymbol{E}_1^T \boldsymbol{E}_1\right)$$

Then, we set the values of η_1 and η_2, which satisfy $m_3 + \eta_1 \le 0$ and $m_5 + \eta_2 \le 0$.

Moreover, we have:

$$\begin{aligned}
\dot{V} &\le \left(m_1 + \tfrac{1}{2} m_2 \hat{A}\right) \boldsymbol{E}_1^T \boldsymbol{E}_1 + \left(m_4 + \tfrac{1}{2} m_2 \hat{A}\right) \boldsymbol{E}_2^T \boldsymbol{E}_2 - m_3 \boldsymbol{\varepsilon}_1^T \boldsymbol{E}_1 + m_5 \left(-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\right)^T \boldsymbol{E}_2 \\
&\quad + \gamma_1 \lambda_1 \|\boldsymbol{E}_1\|_2 tr\left(\bar{\boldsymbol{w}}_1 \hat{\boldsymbol{w}}_1^T\right) + \gamma_2 \lambda_2 \|\boldsymbol{E}_2\|_2 tr\left(\bar{\boldsymbol{w}}_2 \hat{\boldsymbol{w}}_2^T\right)
\end{aligned} \tag{40}$$

Step 1: When condition $V \le V_{\max}$ holds, consider the compact set:

$$\Omega = \Omega_e \times \Omega_d = \left\{\boldsymbol{E}_1^T \boldsymbol{E}_1 + \boldsymbol{E}_2^T \boldsymbol{E}_2 \le 2V_{\max}\right\} \times \{\|x_{1d} + \dot{x}_{1d} + \ddot{x}_{1d}\|_1 \le \chi\}$$

which has a maximum value of $\Omega_{\max}$.

There are constants c and Q that satisfy $m_1 + \frac{1}{2} m_2 \hat{A} \le c$, $m_4 + \frac{1}{2} m_2 \hat{A} \le c$, $\gamma_1 \lambda_1 \|\boldsymbol{E}_1\|_2 \le c$, $\gamma_2 \lambda_2 \|\boldsymbol{E}_2\|_2 \le c$, and $-m_3 \boldsymbol{\varepsilon}_1^T \boldsymbol{E}_1 + m_5 \left(-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\right)^T \boldsymbol{E}_2 = Q$

Hence, we have:

$$\dot{V} \le cV + Q \tag{41}$$

When $V = V_{\max}$ and $\dot{V} \le cV_{\max} + Q$ are satisfied, let c satisfy $c < -Q/2V_{\max}$ and $\dot{V} \le cV + Q < 0$. Thus, according to Eq. (38), there is an invariant set of $V \le V_{\max}$; for all $t > 0$, we have $V \le V_{\max}$. All error signals are semiglobally uniformly bounded in the compact set Ω_e:

$$\Omega_e = \left\{(\boldsymbol{E}_1(i), \boldsymbol{E}_2(i), \boldsymbol{\varepsilon}_1(i), \boldsymbol{\varepsilon}_2(i)) : V \le \frac{Q}{2V_{\max}}\right\} \qquad t \to \infty, \qquad i \quad \text{ergodic } 1, 2, 3.$$

Step 2: According to the properties of the Frobenius norm, we obtain:

$$tr\left(\bar{\boldsymbol{w}}_1 \hat{\boldsymbol{w}}_1^T\right) = tr\left(\bar{\boldsymbol{w}}_1 \left(\bar{\boldsymbol{w}}_1 - \boldsymbol{w}_1\right)^T\right) \le \|\bar{\boldsymbol{w}}_1\|_F^2 - \|\bar{\boldsymbol{w}}_1\|_F \|\boldsymbol{w}_1\|_F$$

$$tr\left(\bar{\boldsymbol{w}}_2 \hat{\boldsymbol{w}}_2^T\right) = tr\left(\bar{\boldsymbol{w}}_2 \left(\bar{\boldsymbol{w}}_2 - \boldsymbol{w}_2\right)^T\right) \le \|\bar{\boldsymbol{w}}_2\|_F^2 - \|\bar{\boldsymbol{w}}_2\|_F \|\boldsymbol{w}_2\|_F$$

$$\|\bar{\boldsymbol{w}}_2\|_F^2 - \|\bar{\boldsymbol{w}}_2\|_F \|\boldsymbol{w}_2\|_F = (\|\bar{\boldsymbol{w}}_2\|_F - \|\boldsymbol{w}_2\|_F)^2 - \frac{1}{2}\|\boldsymbol{w}_2\|_F^2$$

In Eq. (37), we have:

$$\begin{aligned}\dot{V} \leq & \left(m_1 + \tfrac{1}{2}m_2\hat{A}\right)\boldsymbol{E}_1^T\boldsymbol{E}_1 + \left(m_4 + \tfrac{1}{2}m_2\hat{A}\right)\boldsymbol{E}_2^T\boldsymbol{E}_2 - m_3\boldsymbol{\varepsilon}_1^T\boldsymbol{E}_1 + m_5\left(-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\right)^T\boldsymbol{E}_2 \\ & + \eta_1\lambda_1\|\boldsymbol{E}_1\|_2\, tr\left(\bar{\boldsymbol{w}}_1\hat{\boldsymbol{w}}_1^T\right) + \eta_2\lambda_2\|\boldsymbol{E}_2\|_2\, tr\left(\bar{\boldsymbol{w}}_2\hat{\boldsymbol{w}}_2^T\right)\end{aligned} \tag{42}$$

We define the following:

$$H = \dot{V} - \frac{1}{2}m_3^2\|\boldsymbol{\varepsilon}_1\|^2 - \frac{1}{2}m_5^2\|-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\|^2 + \frac{1}{2}\|\boldsymbol{z}_1\|^2 + \frac{1}{2}\|\boldsymbol{z}_2\|^2 \tag{43}$$

We can define the performance index $\boldsymbol{z}$ of Lemma 1: $\boldsymbol{z}_1 = \mathrm{h}_1(\boldsymbol{x}) = \boldsymbol{E}_1$, $\boldsymbol{z}_2 = \mathrm{h}_2(\boldsymbol{x}) = \boldsymbol{E}_2$.
According to Eq. (40), we have:

$$\begin{aligned}H \leq & \left(m_1 + \tfrac{1}{2}m_2\hat{A} + \tfrac{1}{2}\right)\|\boldsymbol{E}_1\|_2^2 + \left(m_4 + \tfrac{1}{2}m_2\hat{A} + \tfrac{1}{2}\right)\|\boldsymbol{E}_2\|_2^2 - m_3\|\boldsymbol{\varepsilon}_1\|_2\|\boldsymbol{E}_1\|_2 + m_5\|-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\|_2\|\boldsymbol{E}_2\|_2 \\ & + \eta_1\lambda_1\|\boldsymbol{E}_1\|_2(\|\bar{\boldsymbol{w}}_2\|_F - \|\boldsymbol{w}_2\|_F)^2 - \tfrac{1}{2}\eta_1\lambda_1\|\boldsymbol{E}_1\|_2\|\boldsymbol{w}_2\|_F^2 + \eta_2\lambda_2\|\boldsymbol{E}_2\|_2(\|\bar{\boldsymbol{w}}_2\|_F - \|\boldsymbol{w}_2\|_F)^2 \\ & - \tfrac{1}{2}\eta_2\lambda_2\|\boldsymbol{E}_2\|_2\|\boldsymbol{w}_2\|_F^2 - \tfrac{1}{2}m_3^2\|\boldsymbol{\varepsilon}_1\|^2 - \tfrac{1}{2}m_5^2\|-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\|^2\end{aligned} \tag{44}$$

To ensure $H \leq 0$, we set:

$$m_4 + \frac{1}{2}m_2\hat{A} + \frac{1}{2} < 0$$
$$m_1 + \frac{1}{2}m_2\hat{A} + \frac{1}{2} < 0$$

and then, we have:

$$\begin{aligned}& -m_3\|\boldsymbol{\varepsilon}_1\|_2\|\boldsymbol{E}_1\|_2 - \frac{1}{2}m_3^2\|\boldsymbol{\varepsilon}_1\|^2 = -\frac{1}{2}(\|\boldsymbol{E}_1\|_2 + m_3\|\boldsymbol{\varepsilon}_1\|_2)^2 - \|\boldsymbol{E}_1\|_2^2 \\ & -m_5\|-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\|_2\|\boldsymbol{E}_2\|_2 - \frac{1}{2}m_5^2\|-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\|^2 = -\frac{1}{2}(\|\boldsymbol{E}_2\|_2 + m_5\|-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\|_2)^2 - \|\boldsymbol{E}_2\|_2^2 \\ & \eta_1\lambda_1\|\boldsymbol{E}_1\|_2(\|\bar{\boldsymbol{w}}_1\|_F - \|\boldsymbol{w}_1\|_F)^2 - \frac{1}{2}\eta_1\lambda_1\|\boldsymbol{E}_1\|_2\|\boldsymbol{w}_1\|_F^2 - \|\boldsymbol{E}_1\|_2^2 = \|\boldsymbol{E}_1\|_2(\eta_1\lambda_1(\|\bar{\boldsymbol{w}}_1\|_F - \|\boldsymbol{w}_1\|_F)^2 - \frac{1}{2}\eta_1\lambda_1\|\boldsymbol{w}_1\|_F^2 - \|\boldsymbol{E}_1\|_2) \\ & \qquad < 0 \\ & \eta_2\lambda_2\|\boldsymbol{E}_2\|_2(\|\bar{\boldsymbol{w}}_2\|_F - \|\boldsymbol{w}_2\|_F)^2 - \frac{1}{2}\eta_2\lambda_2\|\boldsymbol{E}_2\|_2\|\boldsymbol{w}_2\|_F^2 - \|\boldsymbol{E}_2\|_2^2 = \|\boldsymbol{E}_2\|_2(\eta_2\lambda_2(\|\bar{\boldsymbol{w}}_2\|_F - \|\boldsymbol{w}_2\|_F)^2 - \frac{1}{2}\eta_2\lambda_2\|\boldsymbol{w}_2\|_F^2 - \|\boldsymbol{E}_2\|_2) \\ & \qquad < 0\end{aligned} \tag{45}$$

According to Eq. (45), we set η_1 and η_2 to satisfy the following:

$$\eta_1\lambda_1(\|\bar{\boldsymbol{w}}_1\|_F - \|\boldsymbol{w}_1\|_F)^2 - \frac{1}{2}\eta_1\lambda_1\|\boldsymbol{w}_1\|_F^2 - \|\boldsymbol{E}_1\|_2 < 0$$
$$\eta_2\lambda_2(\|\bar{\boldsymbol{w}}_2\|_F - \|\boldsymbol{w}_2\|_F)^2 - \frac{1}{2}\eta_2\lambda_2\|\boldsymbol{w}_2\|_F^2 - \|\boldsymbol{E}_2\|_2 < 0$$

Reducing the learning rates η_1 and η_2 can make the system more stable, but it will reduce the convergence speed.
Moreover, $\dot{V} \leq \frac{1}{2}m_3^2\|\boldsymbol{\varepsilon}_1\|^2 + \frac{1}{2}m_5^2\|-\boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_3\|^2 - \frac{1}{2}\|\boldsymbol{z}_1\|^2 - \frac{1}{2}\|\boldsymbol{z}_2\|^2$, and according to Lemma 1, J_1 and J_2 satisfy $J_1 \leq m_3^2$ and $J_2 \leq m_5^2$, performance indexes $\boldsymbol{z}_1$ and $\boldsymbol{z}_2$ satisfy the requirements, which means that $\boldsymbol{e}_1$, $\dot{\boldsymbol{e}}_1$, $\boldsymbol{e}_2$, and $\dot{\boldsymbol{e}}_2$ all converge.

4. Simulation

4.1. Supply controller

We need to verify the integrity of the aircraft dynamics simulation and better verify the attitude controller discussed in Section 3. The airspeed controller and position controller of the aircraft need to be added before simulation. To simplify the control model, the XY direction control algorithm is omitted, the roll angle and yaw angle are taken as inputs, and only the height controller is retained. The height is controlled by the pitch angle, and the velocity is controlled by the throttle opening.

Step 1: Height controller design

The error $e_h = h_d - h$ is defined, where h_d is the target height.
The controller is designed as follows:

$$\theta_d = k_{hp}e_h + k_{hd}\dot{e}_h \tag{46}$$

where θ_d is the virtual control law taken as the input of the target pitch angle. In Fig. 2 there are no large angle of attack aerodynamic data. To ensure the authenticity of the simulation, the pitch angle is limited to $8° \sim -8°$ ($0.15 \sim -0.15$ [rad]).

Table 2
UAV data 1.

Parameter	Value
$I_{bxx}\,[\mathrm{kg}\cdot\mathrm{m}^2]$	0.00523
$I_{bxz}\,[\mathrm{kg}\cdot\mathrm{m}^2]$	0.00055
$I_{byy}\,[\mathrm{kg}\cdot\mathrm{m}^2]$	0.02813
$I_{bzz}\,[\mathrm{kg}\cdot\mathrm{m}^2]$	0.02321
$M_b\,[kg]$	1.016
$I_{wxx}\,[\mathrm{kg}\cdot\mathrm{m}^2]$	0.00210
$I_{wyy}\,[\mathrm{kg}\cdot\mathrm{m}^2]$	0.00005
$I_{wzz}\,[\mathrm{kg}\cdot\mathrm{m}^2]$	0.00210
$M_w\,[\mathrm{kg}]$	0.167
c [m]	0.1614

Table 3
UAV data 2.

Sweep angle δ_i	0 [rad] (0°)	0.1745 [rad] (10°)	0.3491 [rad] (20°)	0.5236 [rad] (30°)
b [m]	1.2	1.1818	1.1276	1.0392
$S\,[\mathrm{m}^2]$	0.2191	0.2120	0.2045	0.196

Step 2: Velocity controller design

The error $e_v = V_{ad} - V_a$ is defined, where V_{ad} is the target height.
The controller is designed as follows:

$$\delta_t = k_{vp} e_v + k_{vi} \int e_v dt \tag{47}$$

where δ_t is the signal denoting the throttle.

Because of space limitations, we omit the proof. The basic data of the aircraft are in Tables 2 and 3. I_{bxx}, I_{byy}, and I_{bzz} are the moments of inertia of the airframe, and I_{bxz}, I_{bxy}, and I_{bzy} are the products of inertia of the airframe. The airframe is symmetric about $x_b o_b z_b$, $I_{bxy} = 0$; $I_{bzy} = 0$; I_{wxx}, I_{wyy}, and I_{wzz} are the moments of inertia of the wing, M_w is the wing mass, and M_b is the airframe mass.

4.2. Comparative simulation

In this study, a simulation is used to test how well the proposed control scheme provides control.
The aircraft initial conditions are as follows:

$$\begin{bmatrix} u & v & w \end{bmatrix}^T = \begin{bmatrix} 15 & 0 & 0 \end{bmatrix}^T, \quad \begin{bmatrix} \phi & \theta & \varphi \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T, \quad \begin{bmatrix} x & y & h \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T, \quad \begin{bmatrix} p & q & r \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T.$$

The target roll angle is:

$$\phi_d = \begin{cases} 0.4\sin(t)\ [\mathrm{rad}] & 0 \le t \le 60 \\ 0.4\sin(t) - 0.4\ [\mathrm{rad}] & 60 \le t \le 100 \end{cases}$$

The target yaw angle is:

$$\varphi_d = \begin{cases} -0.2\sin(t)\ [\mathrm{rad}] & 0 \le t \le 50 \\ -0.2\sin(t)^\circ - 0.2\ [\mathrm{rad}] & 50 \le t \le 100 \end{cases}$$

The target altitude is $h = 50$ m, and the target speed is $V_a = 20$ m/s. To overcome the complexity explosion caused by $\dot{\boldsymbol{x}}_{2d}$, we use a high pass filter $G_3 = \frac{50s}{s+50}$ instead of differential filter.

The aircraft starts to change from 0 to 0.52 [rad](30°) after 50 s. After 80 s, one of the wings off the aircraft is restored from 0.52 [rad](30°) to 0° due to a failure of the aircraft deformation mechanism. The wing fault on one side will not only lead to a change in the moment of inertia, but also to a change in the aerodynamic parameters. Due to the asymmetry of the aircraft, the aerodynamic parameters C_{l0} and C_{n0} are not 0, and we set $C_{l0} = 0.002$ and $C_{n0} = 0.005$. Sinusoidal targets can continuously incentivize the aircraft, and they can show that the controller can still maintain good control performance when the aircraft makes a large number of maneuvers, meeting the requirements of [47] [48].

In this study, the controller simulation parameters are set as follow: $c_1 = 0.0333$, $k_1 = 1$, $\gamma_1 = 0.0741$, $\eta_1 = 15$, $\lambda_1 = 0.001$, $c_2 = 0.0667$, $k_2 = 1$, $\gamma_2 = 0.5$, $\eta_2 = 20$, and $\lambda_2 = 0.001$. In the velocity controller, $k_{vp} = 1.5$ and $k_{vi} = 0.15$, and in the height controller, $k_{hp} = 0.8$ and $k_{hd} = 0.2$. In the RDLS, the damping factor is $\boldsymbol{\Lambda}(k) = 0.9995$. The method proposed in this paper is represented by f1; f2 uses the gradient descent method instead of RDLS and f3 defines the damping factor as $\boldsymbol{\Lambda}(k) = 1$. We define the root mean square error of the attitude angle as $RMSE = \|\boldsymbol{e}_1\|_2$, and the simulation results are shown as follows.

The results of the simulation are shown in Figs. 8–16. Figs. 8–11 show the comparison of the aircraft attitude angle and reference signal. Fig. 10 and Fig. 12 illustrate the altitude and velocity response. Fig. 13 and Fig. 14 reveal the tracking errors and the root square error (RSE). The aircraft angular rate is shown in Fig. 15. Fig. 16 shows the results of the partial aerodynamic parameters. The f2 method cannot estimate the parameters, and thus we omit it in Fig. 16.

In the first 50 s, the sweep angle of the aircraft is 0°, and the aircraft first climbs and then changes to horizontal flight. Figs. 8–11 show that f1, f2, and f3 track the target trajectory better when the aircraft sweep angle is 0°. The f1 and f3 methods require time to adjust

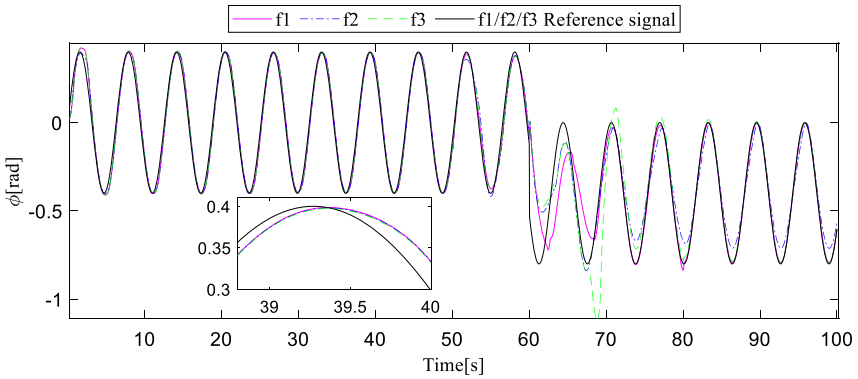


Fig. 8. Roll angle tracking.

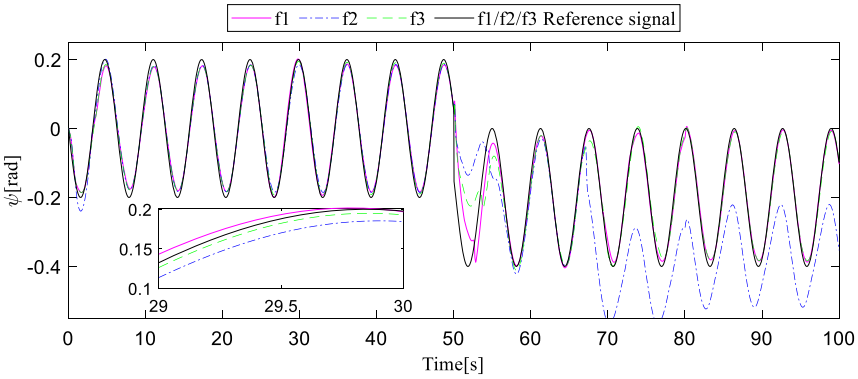


Fig. 9. Yaw angle tracking.

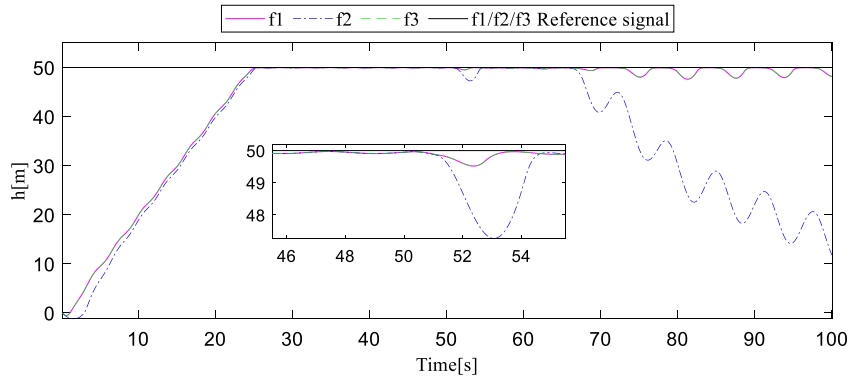


Fig. 10. Altitude tracking.

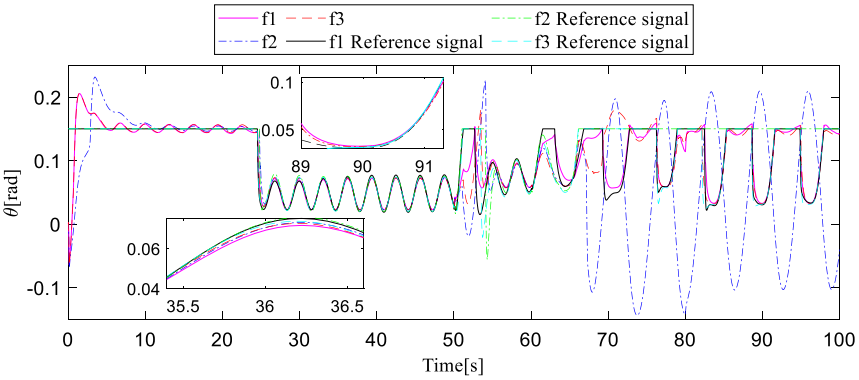


Fig. 11. Pitch angle tracking.

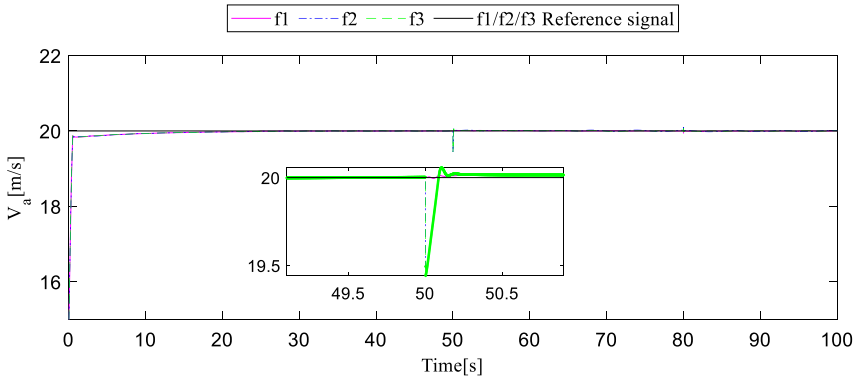


Fig. 12. Velocity tracking.

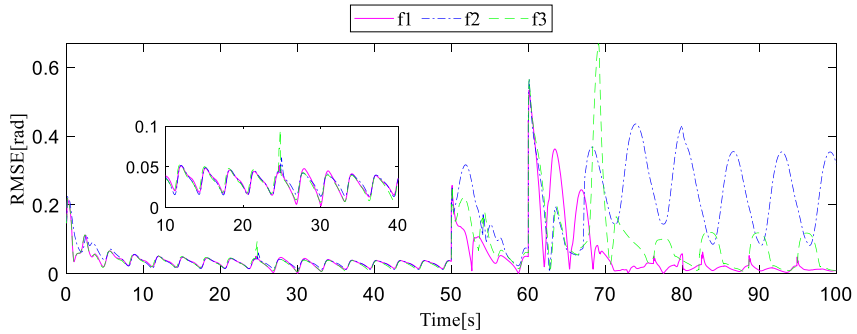


Fig. 13. RMSE.

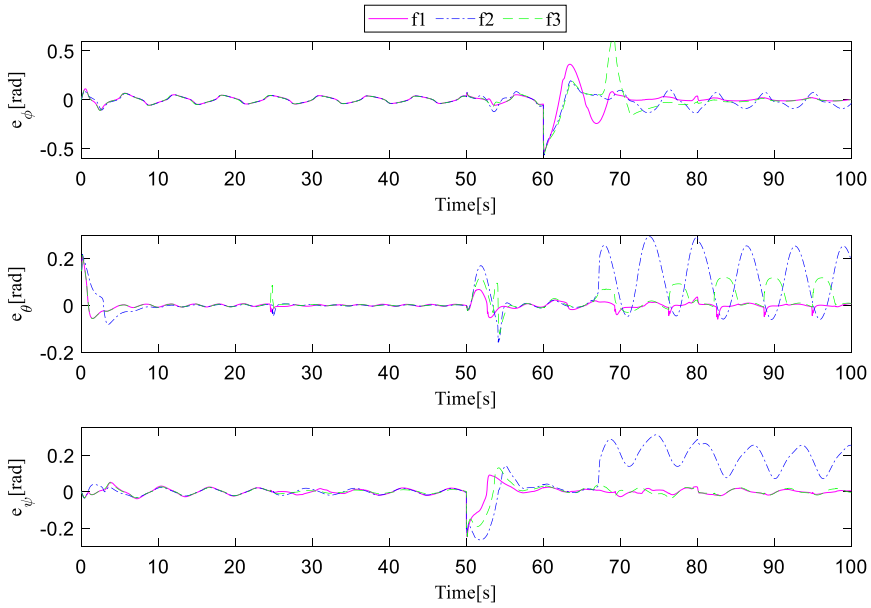


Fig. 14. Attitude angle tracking error.

the parameters, so there is some overshoot in the tracking of the roll angle and pitch angle. Fig. 8 shows that f1 and f3 are better than f2 in tracking the roll angle. In Fig. 9, f1, f2, and f3 can also accurately track the yaw angle. In Fig. 11, in the process of climbing, the changes in the pitch angles of f1 and f3 are smaller than that of f2. After returning to horizontal flight, the changes in altitude of f1 and f3 are smaller than that of f2, so the reference signal required for pitch angle is smaller than that of f2. In terms of error f1 and f3 are better than f2, as shown in Fig. 13 and Fig. 14.

At 50 s, the aircraft sweep angle begins to vary, and the yaw angle command starts to change. At this time, the single neural network in f2 cannot better compensate for $\boldsymbol{f}(\boldsymbol{x})$ in Eq. (14) and the gradient descent method also cannot identify the parameters. Therefore, the roll angle, height, and velocity have chattering phenomenon in Figs. 8–11. This chattering phenomenon is more obvious in the roll angle in Fig. 8 because the lateral flight quality is poor due to the increase in sweep angle. At 60 s, the roll angle command begins to change, and at this time, the tracking roll angle of f2 has not converged, resulting in the aircraft being unable to track the pitch angle and losing

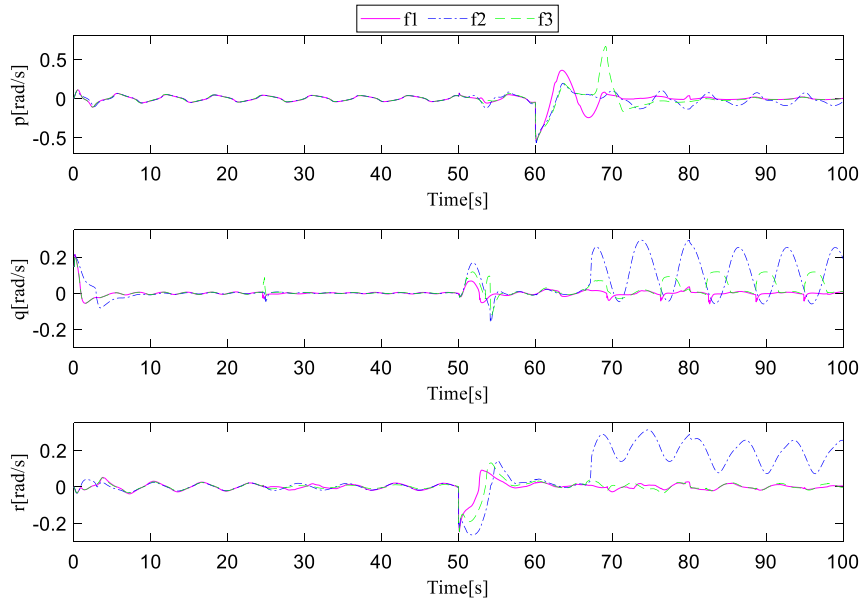


Fig. 15. Angular rate responses.

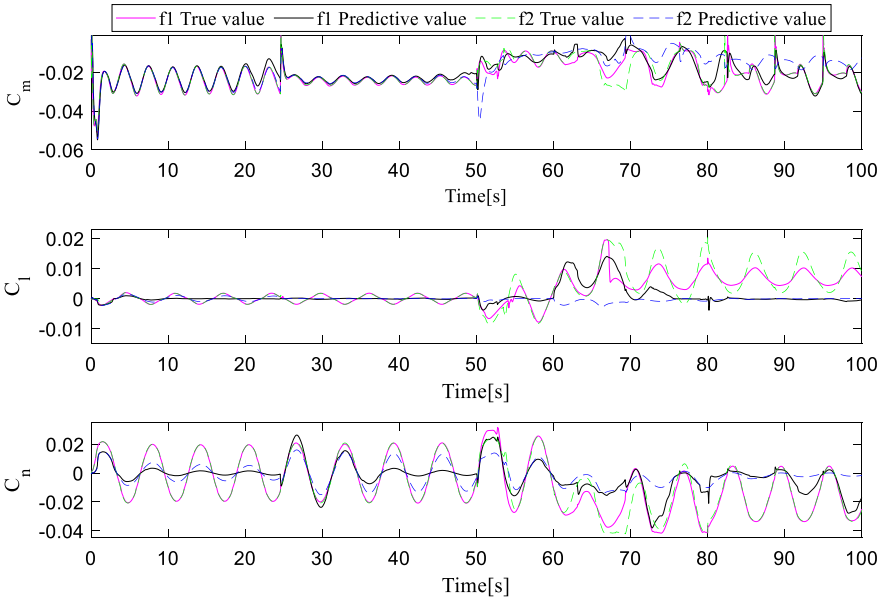


Fig. 16. Partial prediction results.

height control. The methods of f1 and f3 also require a period of time to adjust the roll angle after 60 s, and they can better track the roll angle after 70 s. However, due to the large rolling range, no matter what method is used, the height drops more than before, as shown in Fig. 13.

After 80 s, the variant actuator of the aircraft's right-wing failed and could not be controlled, and the left-wing sweep angle returned to 0°. At this time, the mass and moment of inertia of the aircraft were not symmetrical, and the geometric shape of the aircraft was not asymmetric, resulting in C_{l0} and C_{n0} not being 0 in Eq. (25). At 80 s, the attitude angle had a small jitter. In Fig. 16, f1 and f3 can track C_m accurately, while for C_l and C_n, because there is more coupling in the roll and yaw, insufficient incentive, and a large difference in the magnitudes of the parameters to be identified, the identification of C_l and C_n is not very accurate, and f3 needs a longer time to adapt to sudden changes in the aircraft parameters than f1, which is evident in Fig. 18 and the attitude angle tracking error. In Figs. 8–11, f1 and f3 can still track the attitude angle effectively after 80 s. In Fig. 9, due to the roll angle and yaw angle changing rapidly, the side slip angle is large, and this large side slip angle reduces the lift coefficient, which is the main reason for the aircraft's falling height at this stage. The aircraft can still quickly return to the target height after a small loss of height. Fig. 16 shows that f1 can track the changing parameters faster than f3. Both f1 and f3 can control morphing aircraft in complex situations, but the f1 attitude angle tracking error is smaller than that of f3, as shown in Fig. 13 and Fig. 14.

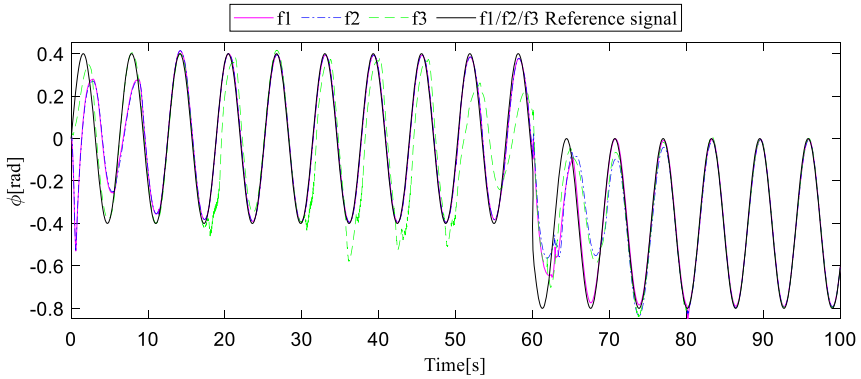


Fig. 17. Roll angle tracking.

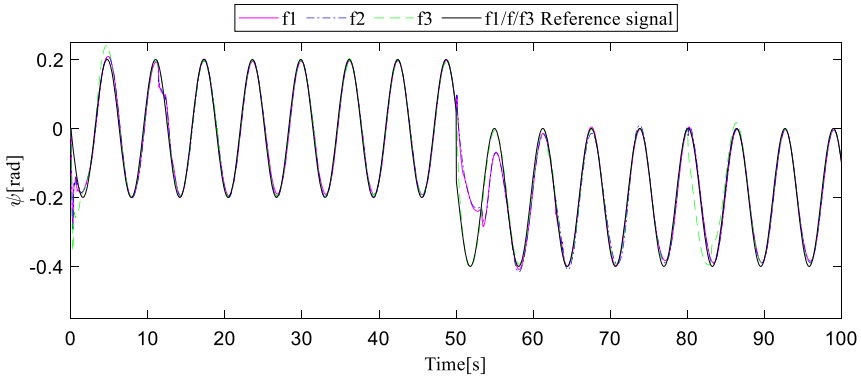


Fig. 18. Yaw angle tracking.

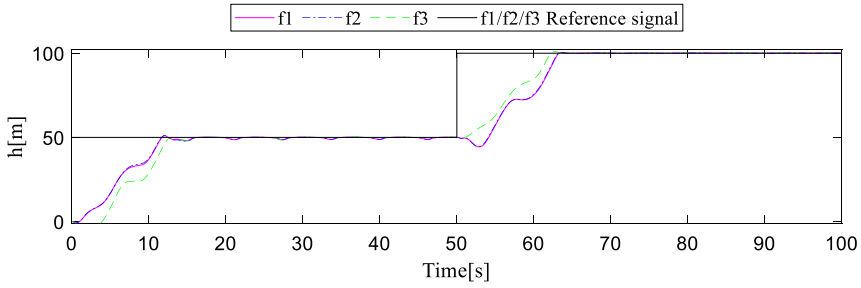


Fig. 19. Altitude tracking.

4.3. Simulation under wind

With the above initial conditions unchanged, a 7 m/s wind is added in both the X and Y directions in the inertial coordinate system, and the target roll angle, yaw angle, and target speed are the same as in Section 4.2. The method proposed in this paper is represented by f1, where f2 define the damping factor $\mathbf{\Lambda}(k) = 1$ and f3 use the normal RBFNN sliding mode.

The target altitude is:

$$h_d = \begin{cases} 50\,[\mathrm{m}] & 0 \le t \le 50 \\ 100\,[\mathrm{m}] & 50 \le t \le 100 \end{cases}$$

The results of the simulation are shown in Figs. 17–24. Figs. 17–20 show the comparison of the aircraft attitude angle and the reference signal. Fig. 19 and Fig. 21 illustrate the altitude and velocity response. Fig. 22 and Fig. 23 reveal the tracking errors and the root means square error (RMSE). Fig. 24 shows the results of the partial aerodynamic parameters.

At the first 50 s, the sweepback angle of the aircraft is 0°, has a large wingspan, and the external wind has lateral and longitudinal disturbances on the aircraft. Large wingspan and lateral wind will cause obvious overshoot of roll angle and yaw angle response in Fig. 1and Fig. 18. The longitudinal disturbance is reflected in the relative velocity with the wind, which is equivalent to bringing an additional velocity to the aircraft. It is obvious from Fig. 19 that the aircraft climbing time is less than that in Fig. 10. Compared with the methods of f1, f2, and f3, f1 and f2 are not significantly different, but they are different from f3 significantly. The overshoot of f3 is small because methods f1 and f2 require time to adjust parameters. Fig. 22 and Fig. 23 show that after 15s, f3 will have a chattering phenomenon, while the attitude angel errors of f1 and f2 are smaller than f3.

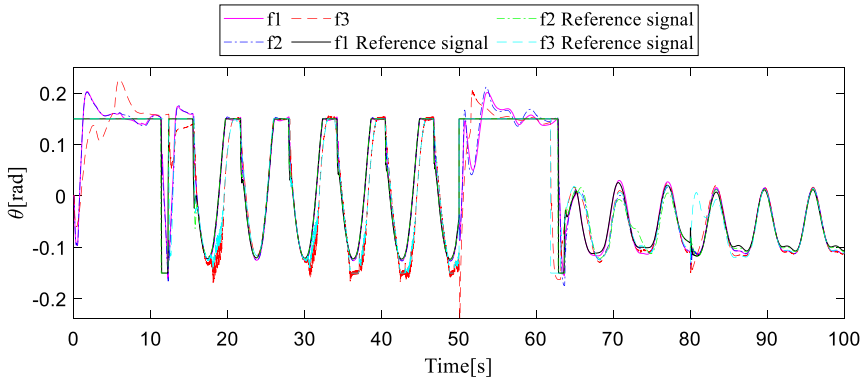


Fig. 20. Pitch angle tracking.

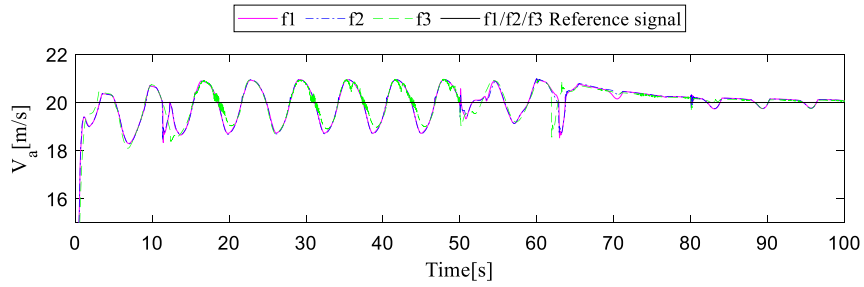


Fig. 21. Velocity tracking.

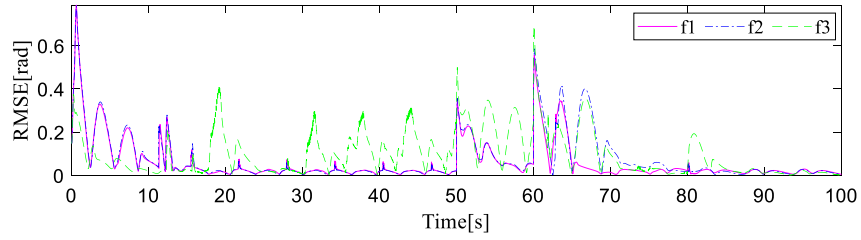


Fig. 22. RMSE.

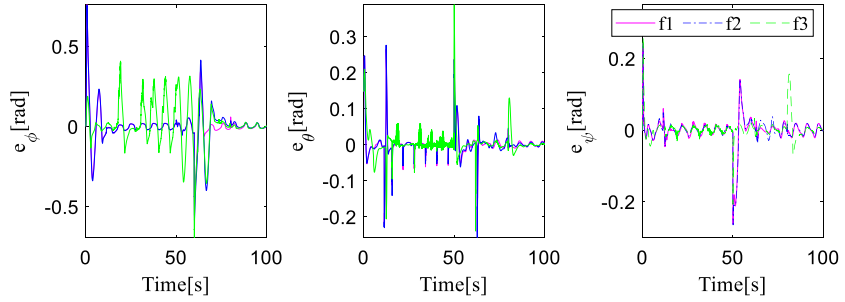


Fig. 23. Attitude angle tracking error.

At 50 s, the aircraft starts to change the sweepback angle and the target heading angle starts to change. Although this leads to poor stability, it increases maneuvering flight capability and makes it possible to better track the attitude angle command. At 60 s, the target roll angle starts to change. In Fig. 22 and Fig. 23 show that the tracking errors of f1 are less than f2 and f3. After 80s, the aircraft has failed as same as in Section 4.2. f1 can still keep the small error in Fig. 22 and Fig. 23. In general, compared with the RBFNN sliding mode control in f3, the method in this paper can still track commands in the wind field.

5. Summary

In this paper, we study the modeling and fault-tolerant control of morphing aircraft. First, a mathematical model of a morphing aircraft is established based on reasonable assumptions. The model describes the changes in the aerodynamic parameters, mass, and moment of inertia of an aircraft at different sweep angles. To address the situations in which the variant mechanism of morphing aircraft is prone to damage and large-angle maneuvers, a passive fault-tolerant attitude controller is constructed. An RBFNN and RDLS are used to

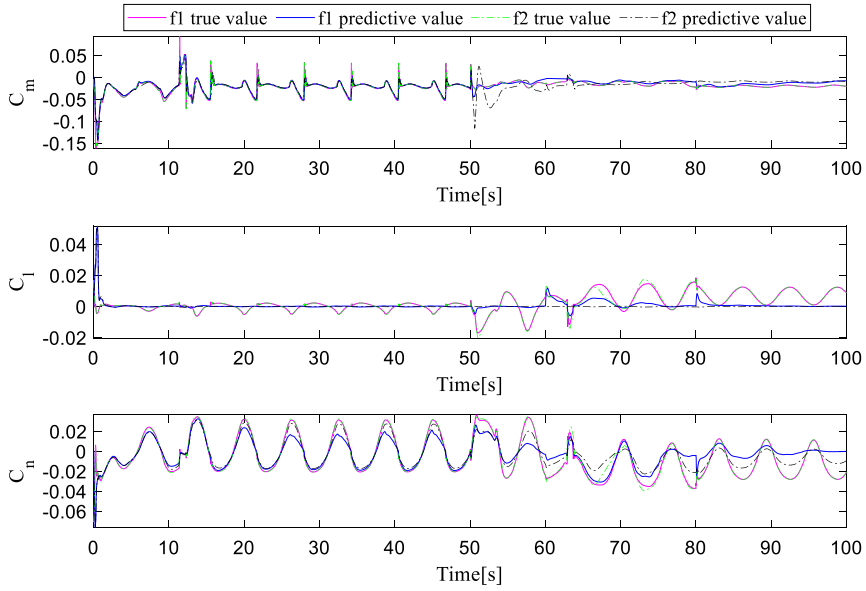


Fig. 24. Partial prediction results.

compensate for the disturbance of the aircraft in real time. More importantly, this makes the controller design independent of the system model. A compact set and Lyapunov function are used to prove its stability, and the L_2 gain is introduced to prove that it can offset the error caused by the RBFNN and RDLS approximation models. The simulation results show that the controller has good performance in large angle rapid maneuvers and can still complete the flight mission well in the case of a fault.

However, this method also has some limitations. First, when the aircraft starts to vary or fails, it needs to give an incentive signal, so that the RBFNN and RDLS can better compensate. Second, for large aircraft, the initial moment of inertia should be known as precisely as possible to prevent the compensation distortion caused by a large order difference in the parameters to be identified. Finally, if the formula of assumption 1 is adopted, the difference in the mass distribution on both sides of the aircraft should not be too large.

In future work, we aim to design committed to designing a more accurate observer to calculate the disturbance and time-varying system model to better compensate for the varying parameters of aircraft, and we will apply it to large flexible aircraft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Appendix A

The detailed expressions of the matrices in the complete equations of motion in Eq. (14) are presented as follows.

$$\boldsymbol{A}(\dot{\boldsymbol{x}}_1) = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix}$$

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \mathcal{T}_1 pq - \mathcal{T}_2 qr + \frac{1}{2}\rho V_a^2 \left(C_p(\alpha,\beta) + C_p^p \frac{b}{2V_a} p + C_p^r \frac{b}{2V_a} r \right) \\ \mathcal{T}_5 \left(r^2 - p^2\right) - \mathcal{T}_6 \left(I_{zz} - I_{xx}\right) pr + \frac{1}{2I_{yy}} \rho V_a^2 \left(C_m(\alpha,\beta) + C_{\mathrm{q}}^m \frac{c}{2V_a} q \right) \\ \mathcal{T}_7 pq - \mathcal{T}_1 qr + \frac{1}{2}\rho V_a^2 \left(C_r(\alpha,\beta) + C_r^p \frac{b}{2V_a} p + C_r^r \frac{b}{2V_a} r \right) \end{bmatrix}$$

$$\boldsymbol{g} = \begin{bmatrix} C_p^{\delta_a} & 0 & C_p^{\delta_r} \\ 0 & C_m^{\delta_e} & 0 \\ C_r^{\delta_a} & 0 & C_r^{\delta_r} \end{bmatrix}$$

References

[1] D.C. Li, S.W. Zhao, A. Da Ronch, J.W. Xiang, J. Drofelnik, Y.C. Li, L. Zhang, Y.N. Wu, M. Kintscher, H.P. Monner, A. Rudenko, S.J. Guo, W.L. Yin, J. Kirn, S. Storm, R.D. Breuker, A review of modelling and analysis of morphing wings, Prog. Aeronaut. Sci. 100 (2018) 46–62.

[2] M. Nemati, A. Jahangirian, Robust aerodynamic morphing shape optimization for high-lift missions, Aerosp. Sci. Technol. 103 (2020) 105897.

[3] F. Nicassio, G. Scarselli, F. Pinto, F. Ciampa, O. Iervolino, M. Meo, Low energy actuation technique of bistable composites for aircraft morphing, Aerosp. Sci. Technol. 75 (2018) 35–46.
[4] Z. Hui, Y. Zhang, G. Chen, Aerodynamic performance investigation on a morphing unmanned aerial vehicle with bio-inspired discrete wing structures, Aerosp. Sci. Technol. 95 (2019) 105419.
[5] T. Yue, X.Y. Zhang, L.X. Wang, J.Q. Ai, Flight dynamic modeling and control for a telescopic wing morphing aircraft via asymmetric wing morphing, Aerosp. Sci. Technol. 70 (2017) 328–338.
[6] T.A. Weisshaar, Morphing aircraft systems: historical perspectives and future challenges, J. Aircr. 50 (2) (2013) 337–353.
[7] R.M. Ajaj, M.I. Friswell, E.I. Saavedra Flores, A. Keane, A.T. Isikveren, G. Allegri, S. Adhikari, An integrated conceptual design study using span morphing technology, J. Intell. Mater. Syst. Struct. 25 (8) (2014) 989–1008.
[8] M. Mohamed, G. Madhavan, Reduced order model based flight control system for a flexible aircraft, IFAC-PapersOnLine 53 (1) (2020) 75–80.
[9] T. Yue, L.X. Wang, J.Q. Ai, Gain self-scheduled H ∞ control for morphing aircraft in the wing transition process based on an LPV model, Chin. J. Aeronaut. 26 (4) (2013) 909–917.
[10] W.L. Jiang, K.S. Wu, Z.L. Wang, Y.N. Wang, Gain-scheduled control for morphing aircraft via switching polytopic linear parameter-varying systems, Aerosp. Sci. Technol. 107 (2020) 106242.
[11] H.Y. Cheng, C.Y. Dong, W.L. Jing, Q. Wang, Y.Z. Hou, Non-fragile switched H_∞ control for morphing aircraft with asynchronous switching, Chin. J. Aeronaut. 30 (3) (2017) 1127–1139.
[12] Q. Wu, Y. Tan, Y. Zhao, et al., Design of integral sliding mode controller with fuzzy switching gain adjustment for morphing aircraft, in: 2021 China Automation Congress (CAC), IEEE, 2021, pp. 1245–1250.
[13] Q. Wu, Y. Tan, Y. Zhao, R. Kang, Q. Xu, T.Y. Jiang, Modeling and switching adaptive control for nonlinear morphing aircraft considering actuator dynamics, Aerosp. Sci. Technol. (2022) 107349.
[14] Z.H. Wu, J.C. Lu, Q. Zhou, J.P. Shi, Modified adaptive neural dynamic surface control for morphing aircraft with input and output constraints, Nonlinear Dyn. 87 (4) (2017) 2367–2383.
[15] C.Y. Bao, P. Wang, G.J. Tang, Integrated method of guidance, control and morphing for hypersonic morphing vehicle in glide phase, Chin. J. Aeronaut. 34 (5) (2021) 535–553.
[16] D. Xu, Z. Hui, Y.Q. Liu, G. Chen, Morphing control of a new bionic morphing UAV with deep reinforcement learning, Aerosp. Sci. Technol. 92 (2019) 232–243.
[17] H.J. Koolstra, J.A. Mulder, Effective model size for the prediction of the lateral control envelope of damaged aircraft, in: AIAA Modeling and Simulation Technologies Conference, 2015, p. 2036.
[18] S. Yin, B. Xiao, S.X. Ding, D.H. Zhou, A review on recent development of spacecraft attitude fault tolerant control system, IEEE Trans. Ind. Electron. 63 (5) (2016) 3311–3320.
[19] Y. Hitachi, H. Liu, Robust thrust-only control of damaged aircraft with horizontal tail loss, in: AIAA Guidance, Navigation, and Control Conference, 2010, p. 7856.
[20] S. Ifqir, C. Combastel, A. Zolghadri, G. Alcalay, P. Goupil, S. Merlet, Fault tolerant multi-sensor data fusion for autonomous navigation in future civil aviation operations, Control Eng. Pract. 123 (2022) 105132.
[21] W. Fan, B. Xu, Y.B. Zhang, S.X. Tang, C.L. Xiang, Adaptive fault-tolerant control of a novel ducted-fan aerial robot against partial actuator failure, Aerosp. Sci. Technol. (2022) 107371.
[22] Y.S. Liu, S. Hong, E. Zio, J.W. Liu, Integrated fault estimation and fault-tolerant control for a flexible regional aircraft, Chin. J. Aeronaut. 35 (3) (2022) 390–399.
[23] C.M. Kaneba, X.W. Mu, X. Li, X.H. Wu, Event triggered control for fault tolerant control system with actuator failure and randomly occurring parameter uncertainty, Appl. Math. Comput. 415 (2022) 126714.
[24] S. Paul, F. Hole, A. Zytek, C.A. Varela, Wind-aware trajectory planning for fixed-wing aircraft in loss of thrust emergencies, in: 2018 IEEE/AIAA 37th Digital Avionics Systems Conference (DASC), IEEE, 2018, pp. 1–10.
[25] G. Shah, Aerodynamic effects and modeling of damage to transport aircraft, in: AIAA Atmospheric Flight Mechanics Conference and Exhibit, 2008, p. 6203.
[26] N. Nguyen, K. Krishnakumar, J. Kaneshige, P. Nespeca, Flight dynamics and hybrid adaptive control of damaged aircraft, J. Guid. Control Dyn. 31 (3) (2008) 751–764.
[27] H. Gao, W. He, Y.M. Zhang, C.Y. Sun, Adaptive finite-time fault-tolerant control for uncertain flexible flapping wings based on rigid finite element method, IEEE Trans. Cybern. 52 (9) (2021) 9036–9047.
[28] Z. Li, X. Chen, M.Y. Xie, Z.H. Zhao, Adaptive fault-tolerant tracking control of flying-wing unmanned aerial vehicle with system input saturation and state constraints, Trans. Inst. Meas. Control 44 (4) (2022) 880–891.
[29] J.J. Ma, C.Peng, Adaptive model-free fault-tolerant control based on integral reinforcement learning for a highly flexible aircraft with actuator faults, Aerosp. Sci. Technol. 119 (2021) 107204.
[30] X.R. Wang, S.H. Sun, Incremental fault-tolerant control for a hybrid quad-plane UAV subjected to a complete rotor loss, Aerosp. Sci. Technol. (2021) 107105.
[31] L. Zhaoying, S. Shuai, L. Hao, Trajectory tracking of tail-sitter aircraft by $\mathcal{L}_1$ adaptive fault tolerant control, J. Syst. Eng. Electron. 32 (6) (2021) 1477–1489.
[32] M. Hosseinzadeh, I. Kolmanovsky, S. Baruah, B. Sinopoli, Reference Governor-based fault-tolerant constrained control, Automatica 136 (2022) 110089.
[33] S.Q. Liu, J.F. Whidborne, Observer-based incremental backstepping sliding-mode fault-tolerant control for blended-wing-body aircrafts, Neurocomputing 464 (2021) 546–561.
[34] J. Wang, S.P. Wang, X.J. Wang, C. Shi, M.M. Tomovic, Active fault tolerant control for vertical tail damaged aircraft with dissimilar redundant actuation system, Chin. J. Aeronaut. 29 (5) (2016) 1313–1325.
[35] S. Ijaz, L. Yan, M.T. Hamayun, C. Shi, Active fault tolerant control scheme for aircraft with dissimilar redundant actuation system subject to hydraulic failure, J. Franklin Inst. 356 (3) (2019) 1302–1332.
[36] D. Zhao, H. Yang, B. Jiang, L.Y. Wen, Attitude stabilization of a flexible spacecraft under actuator complete failure, Acta Astronaut. 123 (2016) 129–136.
[37] N.U. Rahman, J.F. Whidborne, Propulsion and flight controls integration for a blended-wing-body transport aircraft, J. Aircr. 47 (3) (2010) 895–903.
[38] M.J.T. Kammegne, R.M. Botez, L.T. Grigorie, M. Mamou, Y. Mébarki, Proportional fuzzy feed-forward architecture control validation by wind tunnel tests of a morphing wing, Chin. J. Aeronaut. 30 (2) (2017) 561–576.
[39] J.H.S. Fincham, M.I. Friswell, Aerodynamic optimisation of a camber morphing aerofoil, Aerosp. Sci. Technol. 43 (2015) 245–255.
[40] P. Panagiotou, S. Antoniou, K. Yakinthos, Cant angle morphing winglets investigation for the enhancement of the aerodynamic, stability and performance characteristics of a tactical Blended-Wing-Body UAV, Aerosp. Sci. Technol. 123 (2022) 107467.
[41] B.B. Yan, Y. Li, P. Dai, S.X. Liu, Aerodynamic analysis, dynamic modeling, and control of a morphing aircraft, J. Aerosp. Eng. 32 (5) (2019) 04019058.
[42] B.B. Yan, P. Dai, R.F. Liu, M.Z. Xing, S.X. Liu, Adaptive super-twisting sliding mode control of variable sweep morphing aircraft, Aerosp. Sci. Technol. 92 (2019) 198–210.
[43] D. Communier, M.F. Salinas, O.C. Moyao, R.M. Botez, Aero structural modeling of a wing using CATIA V5 and XFLR5 software and experimental validation using the Price-Païdoussis wing tunnel, in: AIAA Atmospheric Flight Mechanics Conference, 2015, p. 558.
[44] A.J. Van Der Schaft, L_2-gain analysis of nonlinear systems and nonlinear state feedback H_∞ control, IEEE Trans. Autom. Control 37 (6) (1992) 770–784.
[45] J. Park, I.W. Sandberg, Universal approximation using radial-basis-function networks, Neural Comput. 3 (2) (1991) 246–257.
[46] J.T. Xu, N.G. Cui, J.C. Tian, H.L. Liu, Study on dynamic decoupling design by using in a method for cruise missile, in: 2008 2nd International Symposium on Systems and Control in Aerospace and Astronautics, IEEE, 2008, pp. 1–5.
[47] T. Zhang, C.C. Li, D.Y. Ma, X.D. Wang, C.Y. Li, An optimal task management and control scheme for military operations with dynamic game strategy, Aerosp. Sci. Technol. 115 (2021) 106815.
[48] S. Chen, Y. Yang, D.Y. Ma, X.D. Wang, K.B. Li, C.Y. Li, Cooperative Guidance Law with Impact Angle Coordination: A Nash Approach, IEEE Trans. Aerosp. Electron. Syst. (2022), https://doi.org/10.1109/TAES.2021.3134998.